

Panel Data Econometrics

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Notation and Scope

We use the baseline panel-data notation

$$y_{it} = x'_{it}\beta + \alpha_i + \lambda_t + u_{it},$$

where i indexes units and t indexes time. Lecture 1 develops the core static panel-data framework, including inference and practical guidance for applied panel work. Lecture 2 then extends that framework to difference-in-differences, random effects, and dynamic panels.

Lecture 1: Foundations

Why Panel Data?

Empirical work is always built on comparisons. The central practical question is: *who is being compared to whom?* A cross-section compares one unit to another unit at a given date. A panel adds a second and often more credible comparison: the unit can be compared to itself over time.

Definition 1

A cross-section observes many units once. A time series observes one unit repeatedly. A panel observes many units repeatedly over time.

Typical panel units are workers, firms, households, schools, hospitals, municipalities, countries, or products. Typical time dimensions are years, quarters, months, or days. The notation $i = 1, \dots, N$ indexes units and $t = 1, \dots, T$ indexes time.

The basic econometric appeal. Suppose we want to estimate the effect of a regressor x_{it} on an outcome y_{it} . If we only have a cross-section, then units with high x_i are compared to units with low x_i . But those units may differ in many other ways. With panel data, we can ask a sharper question:

When a given unit changes x_{it} , does y_{it} also change?

That is the beginning of fixed-effects thinking.

Three data structures. It is useful to contrast the three canonical designs:

- **Cross-section:** compare different units at one moment in time.
- **Time series:** compare one aggregate or one unit across many dates.
- **Panel:** compare different units *and* compare the same unit over time.

This extra dimension is not merely a way to get more observations. If the sample has N units and T periods, then a panel has NT rows, but the main value is usually not sample size alone. The main value is that panel data reveal the sources of variation that ordinary cross-sections conceal.

Between variation and within variation. For a generic regressor x_{it} , define the unit mean

$$\bar{x}_i = \frac{1}{T} \sum_{t=1}^T x_{it}$$

and the grand mean

$$\bar{x} = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T x_{it}.$$

Then we can decompose each observation as

$$x_{it} = \bar{x} + (\bar{x}_i - \bar{x}) + (x_{it} - \bar{x}_i). \quad (1)$$

Equation (1) separates three objects:

- the overall mean $\bar{x}$,
- the *between-unit* component $\bar{x}_i - \bar{x}$,
- the *within-unit* component $x_{it} - \bar{x}_i$.

This decomposition is fundamental. A pooled regression generally mixes between and within information. A fixed-effects regression discards the between component and estimates the effect of the within component only. Much of panel econometrics is therefore about deciding which variation is credible for the question at hand.

A simple wage example. Suppose we study the effect of training hours on wages. In a cross-section, workers with more training may earn more because they are better trained, but also because they are more talented, more motivated, or employed at better firms. Denote training by $Train_{it}$ and wages by $Wage_{it}$. A cross-sectional comparison of

$$Wage_i = \beta Train_i + e_i$$

mixes the effect of training with permanent differences across workers.

With panel data, we can observe the same worker repeatedly:

$$Wage_{it} = \beta Train_{it} + \text{worker effect}_i + \text{other shocks}_{it}.$$

If worker ability is roughly constant over the sample period, then repeated observations make it possible to remove that permanent worker effect and compare the worker to herself over time.

Why this matters. In labor, development, health, finance, and public economics, many of the most important omitted variables are persistent. Ability, family background, entrepreneurial talent, patient severity, managerial quality, and local institutional quality often do not move much from one period to the next. Panel data are valuable precisely because they give us a way to deal with those persistent latent differences.

Why the same-unit comparison is often more credible. To see the logic, suppose a unit-specific omitted factor a_i affects both y_{it} and x_{it} . In a pure cross-section, the comparison between a high- x unit and a low- x unit also compares a high- a unit and a low- a unit. That creates omitted-variable bias. In a panel, if a_i is fixed over time, then the comparison of a unit with its own past values holds a_i constant automatically.

This is the single most important reason panel data are so widely used in applied work. They do not magically create exogeneity, but they often move the empirical design from an implausible comparison to a more defensible one.

What panel data help with. There are at least four recurring gains from a panel:

1. **Controlling for unobserved time-invariant heterogeneity.** This is the classic fixed-effects gain.
2. **Studying adjustment and dynamics.** Once we have repeated observations, we can ask how outcomes evolve before and after a change in policy or exposure.
3. **Separating level differences from change over time.** The same regressor can have a strong between relationship and a weak within relationship, or the reverse.
4. **Improving design transparency.** Panel notation makes clear whether identification comes from long-run differences across units, short-run changes within units, or both.

A useful thought experiment. Imagine two cities observed in two years. City A always has higher public spending and higher educational attainment than City B. A cross-section cannot tell whether spending causes education or whether City A simply has deeper underlying advantages. But if City A changes spending sharply while City B does not, and if the underlying city characteristics are stable, the panel comparison becomes more informative because it tracks changes relative to each city's baseline.

Balanced and unbalanced panels. In practice, not every unit is always observed in every period.

Definition 2

A panel is balanced if every unit is observed in the same set of periods. It is unbalanced if some units are missing in some periods.

Balanced panels simplify notation and algebra. Unbalanced panels are common in actual datasets because of entry, exit, attrition, missing records, or survey nonresponse. Most panel estimators can be adapted to unbalanced panels, but missingness can itself become an econometric issue if it is systematic.

What panel data do not solve. It is equally important to be precise about limitations. Panel data help when the confounder is fixed within unit over time. They do *not* automatically solve:

- time-varying omitted variables,
- reverse causality,
- anticipation effects,
- measurement error,
- endogenous sample attrition,
- spillovers across units.

For example, even if hospital fixed effects remove time-invariant hospital quality, a sudden local epidemic can still change both hospital staffing and patient outcomes. That is a time-varying shock, so panel structure alone does not solve the problem.

Common mistake. “Panel data eliminate omitted-variable bias” is too strong. The correct statement is narrower: panel methods can remove omitted variables that are constant within a unit over time. They do not automatically remove omitted variables that move over time.

The strategic value of panel data. The right way to think about panel data is not as a magic technique, but as an opportunity to ask a better comparison question. Instead of comparing different units with different unobserved histories, we compare a unit to its own past while also controlling for common time shocks when needed. That shift in the comparison is the bridge to the unobserved-effects model.

The Unobserved Effects Model

The workhorse model for static panel analysis is the *unobserved effects model*:

$$y_{it} = x'_{it}\beta + \alpha_i + \lambda_t + u_{it}. \quad (2)$$

This equation is simple, but it organizes almost everything that matters in introductory panel econometrics. Each term plays a distinct role:

- y_{it} is the outcome.
- x_{it} is a vector of observed regressors.
- β is the parameter vector of interest.
- α_i is the unit effect, capturing time-invariant heterogeneity.
- λ_t is the time effect, capturing shocks common to all units in period t .
- u_{it} is the idiosyncratic disturbance, varying across both i and t .

Why this is called an unobserved-effects model. The term “unobserved effects” refers to components of the outcome that are not in the observed regressor vector x_{it} but still matter for y_{it} . The most important one is α_i , the unit-specific effect. In applications, α_i may stand for ability, culture, political capacity, managerial talent, soil quality, neighborhood quality, baseline health, or some other latent characteristic that is hard to measure and largely stable over time.

The component λ_t is equally important in many applications. It captures shocks common to all units in a given period: inflation, macroeconomic policy, business-cycle conditions, national legislation, commodity prices, or technology trends. The remaining disturbance u_{it} is then what is left after accounting for observed regressors, permanent unit heterogeneity, and common period shocks.

Conditional mean interpretation. Equation (2) implies the conditional mean

$$\mathbb{E}[y_{it} \mid x_{it}, \alpha_i, \lambda_t] = x'_{it}\beta + \alpha_i + \lambda_t \quad (3)$$

when $\mathbb{E}[u_{it} \mid x_{it}, \alpha_i, \lambda_t] = 0$. This is useful because it says that the model is additive in three layers:

- an observed part, $x'_{it}\beta$,
- a time-invariant unit-specific part, α_i ,
- a common time-specific part, λ_t .

That additive structure is not just convenient notation. It is an empirical claim. It says, for example, that permanent worker ability enters as a level shift, and that aggregate time shocks also enter as level shifts. Much of the value of panel methods comes from the fact that these additive terms can be removed algebraically.

A scalar version helps intuition. To focus on the main issue, temporarily suppress time effects and consider

$$y_{it} = \beta x_{it} + \alpha_i + u_{it}. \quad (4)$$

This specification says that two people with the same observed regressor x_{it} can still differ systematically in their outcomes because they have different α_i . If one worker is permanently more productive than another, then wages differ even when current training, experience, or hours are the same. The econometric problem is that the high-ability worker may also choose different values of x_{it} .

Why α_i is the central object. The main challenge in panel data is rarely the existence of unobserved heterogeneity by itself. The challenge is *correlated* unobserved heterogeneity. If α_i is correlated with the regressor path $(x_{i1}, \dots, x_{iT})$, then ignoring α_i creates omitted-variable bias.

Formally, define the composite error

$$v_{it} = \alpha_i + \lambda_t + u_{it}.$$

Then pooled regression treats v_{it} like an ordinary disturbance. But if

$$\text{Cov}(x_{it}, \alpha_i) \neq 0,$$

the regressor is correlated with the composite error, and ordinary least squares no longer recovers β .

Why this matters. The panel-data problem is usually not that people, firms, or regions differ. Of course they differ. The problem is that these differences are often related to the regressors we care about. Once that happens, an unobserved effect becomes an omitted variable, and the entire question becomes how to handle it.

Examples of correlated unobserved effects. The logic is easier to see in concrete settings:

- **Education and earnings.** More able workers may choose more education. Then ability is part of α_i and is correlated with schooling.
- **Firm investment and cash flow.** More productive firms may have persistently higher cash flow and investment opportunities. Productivity is part of α_i .
- **Municipal spending and outcomes.** Better-run municipalities may both spend more effectively and have better outcomes for reasons not fully observed in the data.

- **Health utilization and insurance.** Sicker individuals may select different insurance plans, so baseline health status belongs in α_i and is correlated with treatment intensity.

Time effects are conceptually different from unit effects. Unit effects α_i control for permanent differences *across units*. Time effects λ_t control for aggregate shocks *across periods*. In many empirical settings both are needed. Suppose we estimate the effect of firm-level leverage on investment. Firm fixed effects remove stable differences in management quality or industry position, but they do nothing about a recession that hits all firms in a given year. That recession belongs in λ_t .

Strict exogeneity. The central exogeneity condition for static fixed-effects analysis is *strict exogeneity*:

$$\mathbb{E}[u_{it} \mid \alpha_i, \lambda_t, x_{i1}, \dots, x_{iT}] = 0 \quad \text{for each } t. \quad (5)$$

This condition is stronger than contemporaneous exogeneity. It does *not* merely require u_{it} to be uncorrelated with current regressors x_{it} . It requires orthogonality to the *entire regressor history and future path* for the unit.

To see why this is demanding, note that (5) implies

$$\mathbb{E}[u_{it} \mid \alpha_i, \lambda_t, x_{it}] = 0$$

but the reverse implication does not hold. A shock today can violate strict exogeneity by changing next period's regressors even if current regressors are uncorrelated with today's shock.

For instance, suppose a positive productivity shock raises current profits, and the firm responds by increasing next period's employment or capital. Then u_{it} helps predict $x_{i,t+1}$, so (5) fails. This is one reason static fixed-effects methods are not automatically valid in dynamic decision settings.

What strict exogeneity buys us. Under (5), once we remove α_i and λ_t , the remaining variation in x_{it} behaves like an exogenous regressor in the transformed equation. This is the condition that justifies fixed-effects estimation in the standard static model.

A covariance-based view. It is often helpful to rewrite the key issue using covariances. Suppressing time effects for clarity, the pooled regression error is

$$v_{it} = \alpha_i + u_{it}.$$

Then consistency of pooled ordinary least squares (OLS) requires

$$\text{Cov}(x_{it}, v_{it}) = \text{Cov}(x_{it}, \alpha_i) + \text{Cov}(x_{it}, u_{it}) = 0.$$

Even if $\text{Cov}(x_{it}, u_{it}) = 0$, pooled OLS still fails whenever $\text{Cov}(x_{it}, \alpha_i) \neq 0$. The panel estimator is therefore designed around one specific task: separate out the persistent component α_i from the transitory component u_{it} .

Two competing philosophies. At this point panel econometrics splits into two broad approaches:

1. **Fixed effects:** allow α_i to be arbitrarily correlated with the regressors and remove it before estimation.
2. **Random effects:** assume α_i is orthogonal to the regressors and use both within and between variation.

The fixed-effects approach is usually more credible in observational work because it does not ask us to believe that permanent latent traits are unrelated to observed choices.

Common mistake. Students often think “fixed effects” means the unit effects are nonrandom constants and “random effects” means they are random variables. That is not the real distinction. The real distinction is whether we allow α_i to be correlated with the regressors. Fixed effects allow that correlation; random effects rule it out.

Three practical questions to ask in every application. Before estimating any panel model, it helps to ask:

1. What is the likely economic content of α_i ?
2. Is that unobserved effect plausibly correlated with the regressors?
3. After removing α_i and perhaps λ_t , what source of variation is left to identify β ?

These questions are more informative than mechanically asking whether a regression has unit dummies.

A final interpretation. Equation (2) is best thought of as a disciplined description of how panel outcomes are generated. It says that observed outcomes come from three layers: observed covariates, stable unit-specific heterogeneity, and time-specific common shocks, plus an idiosyncratic remainder. Once the model is written this way, the next step is immediate: if we ignore α_i , what bias arises, and how

do fixed-effects and first-difference methods remove it? That is the purpose of the next sections.

Why this matters. Removing α_i is algebra. Identifying β is economics. The real question is whether the variation left after removing α_i and perhaps λ_t is plausibly exogenous.

Why Pooled OLS Fails

Suppose we ignore the panel structure and estimate

$$y_{it} = x'_{it}\beta + v_{it}, \quad v_{it} = \alpha_i + \lambda_t + u_{it}. \quad (6)$$

This is pooled OLS. It stacks all NT observations and runs an ordinary regression as if the data were one large cross-section. The key maintained assumption would then be

$$\mathbb{E}[v_{it} \mid x_{it}] = 0.$$

But in panel settings this assumption is usually the first one to fail, because v_{it} contains the unit effect α_i .

What pooled OLS is really doing. Pooled OLS compares:

- one unit to another unit in the same period,
- the same unit to itself over time,
- and all mixtures of those two comparisons.

That is already a warning sign. If the cross-unit comparisons are contaminated by unobserved heterogeneity, then the pooled coefficient combines clean and contaminated sources of variation.

A simple wage-panel example. Let y_{it} be log wages and x_{it} be training hours. Suppose some workers are permanently more able, more motivated, or employed by better firms. Put all of that into α_i . Then the model is

$$y_{it} = \beta x_{it} + \alpha_i + u_{it}.$$

Workers with high α_i may systematically choose more training. A pooled regression then compares high-training workers to low-training workers, but those workers differ not only in training, they also differ in latent ability. The pooled coefficient therefore blends the effect of training with the effect of ability.

Bias from omitting the unit effect. To isolate the main issue, suppress time effects or imagine they have been absorbed by period dummies:

$$y_{it} = \beta x_{it} + \alpha_i + u_{it}. \quad (7)$$

Theorem 1 (Probability limit of pooled OLS)

Consider the scalar model in (7). If $\text{Var}(x_{it}) > 0$, then the pooled OLS estimator satisfies

$$\text{plim } \hat{\beta}^{POLS} = \beta + \frac{\text{Cov}(x_{it}, \alpha_i)}{\text{Var}(x_{it})} + \frac{\text{Cov}(x_{it}, u_{it})}{\text{Var}(x_{it})}. \quad (8)$$

In particular, even if $\text{Cov}(x_{it}, u_{it}) = 0$, pooled OLS is inconsistent whenever $\text{Cov}(x_{it}, \alpha_i) \neq 0$.

Proof 1

For a scalar regressor, the population regression coefficient from regressing y_{it} on x_{it} is

$$\frac{\text{Cov}(x_{it}, y_{it})}{\text{Var}(x_{it})}.$$

Substituting (7) into the numerator gives

$$\begin{aligned} \text{Cov}(x_{it}, y_{it}) &= \text{Cov}(x_{it}, \beta x_{it} + \alpha_i + u_{it}) \\ &= \beta \text{Var}(x_{it}) + \text{Cov}(x_{it}, \alpha_i) + \text{Cov}(x_{it}, u_{it}). \end{aligned}$$

Dividing both sides by $\text{Var}(x_{it})$ yields (8).

The theorem is just omitted-variable bias written in panel notation. The omitted variable is α_i , the permanent component of the unit. In many applications that omitted term contains exactly the economic object we worry about:

- worker ability in earnings equations,
- firm productivity in investment equations,
- patient severity in health-care use,
- municipality capacity in public-finance panels.

Why adding time dummies is not enough. Students often think that adding period dummies already “uses the panel structure.” It does help with common time shocks λ_t , but it does nothing to remove permanent unit heterogeneity α_i . A regression with year dummies but no unit dummies is still comparing different units with different latent traits.

Pooled OLS mixes between and within variation. The deeper issue is not only omitted-variable bias. It is also that pooled OLS uses the wrong source of variation for the question that panel data were meant to answer.

For any variable z_{it} , write

$$z_{it} = \bar{z}_i + (z_{it} - \bar{z}_i),$$

where $\bar{z}_i$ is the unit mean. The first term is the between-unit component and the second term is the within-unit component.

Theorem 2 (Covariance decomposition in a balanced panel)

For any two variables x_{it} and y_{it} observed in a balanced panel,

$$\text{Cov}(x_{it}, y_{it}) = \text{Cov}(\bar{x}_i, \bar{y}_i) + \text{Cov}(x_{it} - \bar{x}_i, y_{it} - \bar{y}_i). \quad (9)$$

Proof 2

Write

$$x_{it} = \bar{x}_i + \tilde{x}_{it}, \quad y_{it} = \bar{y}_i + \tilde{y}_{it},$$

where $\tilde{x}_{it} = x_{it} - \bar{x}_i$ and $\tilde{y}_{it} = y_{it} - \bar{y}_i$. Then

$$\begin{aligned} \text{Cov}(x_{it}, y_{it}) &= \text{Cov}(\bar{x}_i + \tilde{x}_{it}, \bar{y}_i + \tilde{y}_{it}) \\ &= \text{Cov}(\bar{x}_i, \bar{y}_i) + \text{Cov}(\bar{x}_i, \tilde{y}_{it}) + \text{Cov}(\tilde{x}_{it}, \bar{y}_i) + \text{Cov}(\tilde{x}_{it}, \tilde{y}_{it}). \end{aligned}$$

It remains to show that the two cross terms are zero. Because $\bar{x}_i$ is constant over t within unit,

$$\text{Cov}(\bar{x}_i, \tilde{y}_{it}) = \mathbb{E}[\bar{x}_i \tilde{y}_{it}] - \mathbb{E}[\bar{x}_i] \mathbb{E}[\tilde{y}_{it}].$$

Now

$$\mathbb{E}[\bar{x}_i \tilde{y}_{it}] = \mathbb{E} \left[\bar{x}_i \cdot \frac{1}{T} \sum_{t=1}^T (y_{it} - \bar{y}_i) \right] = 0$$

because $\sum_{t=1}^T (y_{it} - \bar{y}_i) = 0$ for every unit. Also $\mathbb{E}[\tilde{y}_{it}] = 0$. Hence $\text{Cov}(\bar{x}_i, \tilde{y}_{it}) = 0$. The same argument shows $\text{Cov}(\tilde{x}_{it}, \bar{y}_i) = 0$. Substituting back proves (9).

Equation (9) is extremely informative. It says that the sample covariance behind pooled OLS has two pieces:

1. a **between** covariance, comparing average outcomes of high- x units to average outcomes of low- x units;
2. a **within** covariance, comparing a unit to its own time average.

If the between component is contaminated by correlation between $\bar{x}_i$ and α_i , pooled OLS is pulled away from the within-unit relationship. So even when panel data are available, pooled OLS often answers the wrong empirical question.

A two-worker illustration. Suppose worker A has high ability and worker B has low ability. Assume training has no causal effect at all, so the true β is zero. But worker A always takes more training and always earns higher wages because of ability. Then a pooled regression will likely estimate a positive relationship between training and wages, even though training is causally irrelevant in this simple story. The regression is not detecting the effect of training. It is detecting that training is more common among high-ability workers.

Now imagine that worker A receives extra training in period 2 while her ability stays fixed. If her wage does not change, the within-worker evidence says the causal effect is close to zero. This is exactly why fixed effects focus on changes within the same unit rather than differences across units.

Why this matters. Pooled OLS is not merely “less sophisticated” than fixed effects. It answers a different question. It combines cross-unit comparisons with within-unit comparisons, and the cross-unit part is often the least credible part in observational data.

What pooled OLS can still be useful for. There are cases where pooled OLS is informative:

- if α_i is genuinely uncorrelated with the regressors,
- if the goal is purely descriptive rather than causal,
- if the panel dimension is very short and within variation is almost absent.

But in most applied panel settings, especially when the objective is causal interpretation, one should begin from the presumption that α_i may be correlated with x_{it} .

Common mistake. Adding many observable controls to pooled OLS does not generally neutralize latent unit heterogeneity. If the remaining unobserved component is still correlated with x_{it} , the pooled estimate remains biased.

Fixed Effects as Within-Unit Identification

Fixed effects solve the omitted-unit-effect problem by changing the comparison. Instead of asking whether high- x units tend to have high y , fixed effects ask whether a given unit tends to have high y in the periods when that same unit has high x .

The core identification idea. Return to the static model without time effects:

$$y_{it} = x'_{it}\beta + \alpha_i + u_{it}. \quad (10)$$

The key maintained restriction is not that units are identical. It is the weaker and more plausible statement that the omitted heterogeneity is *time-invariant*. If α_i does not move over time, then we can eliminate it by subtracting out a unit-specific baseline.

Dummy-variable view. One way to do that is to include a full set of unit indicators:

$$y_{it} = x'_{it}\beta + \sum_{j=1}^{N-1} \delta_j 1\{i = j\} + u_{it}. \quad (11)$$

This is the least-squares dummy-variable representation. It is conceptually transparent: every unit gets its own intercept. In practice it can be cumbersome when N is large, but modern software handles it easily.

Within-transformation view. The same estimator can be derived with more revealing algebra. Take the time average of (10) for unit i :

$$\bar{y}_i = \bar{x}'_i\beta + \alpha_i + \bar{u}_i. \quad (12)$$

Now subtract (12) from (10):

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)'\beta + (u_{it} - \bar{u}_i). \quad (13)$$

The unit effect disappears because $\alpha_i - \alpha_i = 0$.

Define the demeaned variables

$$\check{y}_{it} = y_{it} - \bar{y}_i, \quad \check{x}_{it} = x_{it} - \bar{x}_i, \quad \check{u}_{it} = u_{it} - \bar{u}_i.$$

Then the transformed regression is

$$\check{y}_{it} = \check{x}'_{it}\beta + \check{u}_{it}. \quad (14)$$

Applying OLS to (14) gives

$$\hat{\beta}^{FE} = \left(\sum_{i=1}^N \sum_{t=1}^T \check{x}_{it} \check{x}'_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \check{x}_{it} \check{y}_{it} \right). \quad (15)$$

What this estimator means. Equation (15) makes the identification strategy explicit. Fixed effects use only deviations from a unit's own average:

- if a regressor is above its unit mean in some period, fixed effects ask whether the outcome is also above its unit mean in that period;

- if a regressor never moves within a unit, that unit contributes no identifying information for that regressor.

This is why fixed effects are also called the *within estimator*.

Why this matters. The fixed-effects regression answers a disciplined question: when the same unit changes x_{it} , how does y_{it} change relative to that unit's own baseline? In many applications this is the economically relevant comparison.

A useful picture to keep in mind. Imagine each unit has its own horizontal reference line equal to its average outcome. Fixed effects ask whether periods in which x_{it} is above the unit's usual level are also periods in which y_{it} is above the unit's usual level. The regression is therefore built out of vertical distances from each unit's own mean, not from the overall sample mean.

Theorem 3 (Least-squares dummy variables and within estimation are equivalent)

In the static model (10), the coefficient on x_{it} obtained from the dummy-variable regression (11) is identical to the coefficient obtained from the demeaned regression (14).

Proof 3

Consider the least-squares problem

$$\min_{\beta, \alpha_1, \dots, \alpha_N} \sum_{i=1}^N \sum_{t=1}^T (y_{it} - x'_{it}\beta - \alpha_i)^2.$$

For a fixed value of β , differentiate with respect to α_i :

$$\frac{\partial}{\partial \alpha_i} \sum_{t=1}^T (y_{it} - x'_{it}\beta - \alpha_i)^2 = -2 \sum_{t=1}^T (y_{it} - x'_{it}\beta - \alpha_i).$$

Setting this derivative equal to zero yields

$$\hat{\alpha}_i(\beta) = \bar{y}_i - \bar{x}'_i\beta.$$

Substituting this back into the objective function gives the concentrated least-squares problem

$$\min_{\beta} \sum_{i=1}^N \sum_{t=1}^T [(y_{it} - \bar{y}_i) - (x_{it} - \bar{x}_i)' \beta]^2.$$

But this is exactly the OLS criterion from regressing $\tilde{y}_{it}$ on $\tilde{x}_{it}$. Therefore the slope coefficient from the dummy-variable regression is identical to the slope coefficient from the demeaned regression.

This equivalence matters because it gives us two interpretations at once:

- **Economically:** each unit has its own intercept.
- **Statistically:** we identify β from within-unit covariation only.

Why time-invariant regressors disappear. Suppose z_i does not vary over time. Then

$$z_i - \bar{z}_i = 0.$$

So a fixed-effects regression cannot estimate the effect of regressors such as gender, inherited geography, colonial origin, or any other variable that is constant within unit over the sample.

This is not a flaw in the method. It is the arithmetic consequence of the identifying idea. If a regressor never changes within a unit, there is no within-unit comparison to exploit.

Fixed effects need within variation. The previous point has an important practical implication. A regressor may have a great deal of cross-sectional variation but very little within-unit movement. In that case the pooled regression may look precise while the fixed-effects estimate is imprecise. The reason is not a software problem. The reason is that the data contain little of the variation that fixed effects require.

Common mistake. A fixed-effects coefficient is not a cross-sectional comparison with many controls. It is a within-unit estimate. If there is almost no within-unit movement in a regressor, the fixed-effects estimate will be noisy no matter how large the pooled sample is.

Consistency of fixed effects under strict exogeneity. The transformed equation (14) has removed α_i , but we still need the transformed regressor to be uncorrelated with the transformed error. This is where strict exogeneity enters:

$$\mathbb{E}[u_{it} \mid \alpha_i, x_{i1}, \dots, x_{iT}] = 0 \quad \text{for each } t.$$

Theorem 4 (Orthogonality behind fixed effects)

Under strict exogeneity and finite second moments,

$$\mathbb{E}[\ddot{x}_{it}\ddot{u}_{it}] = 0 \quad \text{for each } t. \tag{16}$$

Consequently, if the matrix $\mathbb{E}[\ddot{x}_{it}\ddot{x}_{it}']$ is nonsingular, the fixed-effects estimator is consistent as $N \rightarrow \infty$ with fixed T .

Proof 4

Because $\ddot{x}_{it}$ is a function of the full regressor path $(x_{i1}, \dots, x_{iT})$,

$$\mathbb{E}[\ddot{x}_{it}u_{is}] = \mathbb{E}[\ddot{x}_{it}\mathbb{E}[u_{is} \mid \alpha_i, x_{i1}, \dots, x_{iT}]] = 0$$

for every $s = 1, \dots, T$ by strict exogeneity. Therefore

$$\mathbb{E}[\ddot{x}_{it}\ddot{u}_i] = \frac{1}{T} \sum_{s=1}^T \mathbb{E}[\ddot{x}_{it}u_{is}] = 0.$$

Taking $s = t$ also gives $\mathbb{E}[\ddot{x}_{it}u_{it}] = 0$. Hence

$$\mathbb{E}[\ddot{x}_{it}\ddot{u}_{it}] = \mathbb{E}[\ddot{x}_{it}(u_{it} - \bar{u}_i)] = \mathbb{E}[\ddot{x}_{it}u_{it}] - \mathbb{E}[\ddot{x}_{it}\bar{u}_i] = 0.$$

This establishes the moment condition in (16). Standard law-of-large-numbers arguments then imply consistency of (15) when the within second-moment matrix is nonsingular.

Why strict exogeneity is stronger than students first think. The condition above rules out feedback from shocks to future regressors. For example, if a good productivity shock today causes a firm to invest more tomorrow, then current u_{it} is correlated with future $x_{i,t+1}$. In that case the fixed-effects estimator in the static model is generally not valid without further structure.

A simple interpretation with one regressor. When x_{it} is scalar, the fixed-effects estimator can be written as

$$\hat{\beta}^{FE} = \frac{\sum_{i=1}^N \sum_{t=1}^T (x_{it} - \bar{x}_i)(y_{it} - \bar{y}_i)}{\sum_{i=1}^N \sum_{t=1}^T (x_{it} - \bar{x}_i)^2}.$$

So $\hat{\beta}^{FE}$ is just the slope from regressing the within-unit deviations of y on the within-unit deviations of x . This formula is often the best one to keep in mind.

A concrete example. Suppose we follow students across semesters and study the effect of study hours on grades. Some students are permanently stronger than others, more organized, or have better prior preparation. Those factors belong in α_i . Fixed effects remove those permanent student traits and estimate the relationship between *changes* in study hours and *changes* in grades for the same student. If Alice always studies more than Bob because she is more disciplined, that permanent difference no longer drives the estimate.

What fixed effects do and do not solve. Fixed effects are powerful, but their scope is exact:

- they remove omitted variables that are constant within unit over time;
- they do not remove omitted variables that change over time within unit;
- they do not solve reverse causality by themselves;
- they do not solve measurement error by themselves.

So the move from pooled OLS to fixed effects is a major improvement, but it is not the end of econometric thinking.

First Differences Versus Fixed Effects

Fixed effects and first differences are close relatives. Both methods start from the same unobserved-effects model,

$$y_{it} = x'_{it}\beta + \alpha_i + u_{it}, \quad (17)$$

and both remove the time-invariant unit effect α_i . The difference lies in *how* they remove it and therefore in *what weighting of the time dimension* they use.

The first-difference transformation. Take the difference between period t and period $t - 1$:

$$\Delta y_{it} = \Delta x'_{it}\beta + \Delta u_{it}, \quad t = 2, \dots, T, \quad (18)$$

where

$$\Delta y_{it} = y_{it} - y_{i,t-1}, \quad \Delta x_{it} = x_{it} - x_{i,t-1}, \quad \Delta u_{it} = u_{it} - u_{i,t-1}.$$

The unit effect disappears because $\alpha_i - \alpha_i = 0$. Algebraically, first differencing is therefore as valid a way to remove α_i as demeaning.

The identification idea is still within-unit. Equation (18) says that the slope β is identified from *changes* within a unit. If a firm's leverage rises from one year to the next, first differences ask whether investment also rises from one year to the next, net of the permanent firm-specific component. The comparison is therefore local in time: today versus yesterday, or this year versus last year.

How first differencing differs conceptually from fixed effects. The within estimator compares each observation to the unit's average over the whole sample. First differencing compares each observation only to the immediately preceding observation. That difference sounds small, but it matters.

- **Fixed effects** ask whether a period is above or below the unit's long-run average.
- **First differences** ask whether the variable rose or fell relative to the previous period.

So the two estimators use the same broad source of variation, but they filter that variation differently.

Example 1

Suppose y_{it} is a student's grade and x_{it} is study hours across four semesters. Fixed effects compare each semester to the student's average semester. First

differences compare semester 2 to semester 1, semester 3 to semester 2, and semester 4 to semester 3. If the student gradually becomes more mature or more efficient at studying, those nearby comparisons may feel more natural than comparing every semester to the overall mean. But if each semester contains a lot of short-run noise, differencing can also make the data noisier.

Moment condition for first differences. The first-difference estimator is obtained by applying OLS to (18). In the scalar-regressor case,

$$\hat{\beta}^{FD} = \frac{\sum_{i=1}^N \sum_{t=2}^T \Delta x_{it} \Delta y_{it}}{\sum_{i=1}^N \sum_{t=2}^T (\Delta x_{it})^2}. \quad (19)$$

Under the standard static-panel strict-exogeneity condition,

$$\mathbb{E}[u_{it} \mid \alpha_i, x_{i1}, \dots, x_{iT}] = 0 \quad \text{for each } t,$$

we also have

$$\mathbb{E}[\Delta x_{it} \Delta u_{it}] = 0.$$

The logic is the same as with fixed effects: because Δx_{it} is a function of the regressor path $(x_{i1}, \dots, x_{iT})$, strict exogeneity implies orthogonality between the transformed regressor and the transformed error.

Theorem 5 (Fixed effects and first differences are identical when $T = 2$)

In the model (17), suppose every unit is observed in exactly two periods. Then the fixed-effects estimator and the first-difference estimator produce the same slope coefficient.

Proof 5

When $T = 2$,

$$\bar{y}_i = \frac{y_{i1} + y_{i2}}{2} \quad \text{and} \quad \bar{x}_i = \frac{x_{i1} + x_{i2}}{2}.$$

Therefore,

$$y_{i2} - \bar{y}_i = \frac{y_{i2} - y_{i1}}{2} = \frac{\Delta y_{i2}}{2}, \quad y_{i1} - \bar{y}_i = -\frac{y_{i2} - y_{i1}}{2} = -\frac{\Delta y_{i2}}{2}.$$

The same identities hold for x_{it} :

$$x_{i2} - \bar{x}_i = \frac{\Delta x_{i2}}{2}, \quad x_{i1} - \bar{x}_i = -\frac{\Delta x_{i2}}{2}.$$

Hence every demeaned observation is just a constant multiple of the corresponding differenced observation. In the fixed-effects formula, both the numerator and denominator are therefore multiplied by the same constant factor relative to the first-difference formula. The slope coefficients are identical.

The theorem is worth remembering because it tells us something deeper than a special-case identity. With only two periods, there is essentially just *one within-unit movement* available. Fixed effects and first differences are merely two notational ways of extracting it.

Why the estimators differ once $T > 2$. For more than two periods, the equality in Theorem 5 no longer holds. The reason is that the within transformation uses all periods simultaneously, whereas first differencing uses only adjacent comparisons. For $T = 4$, for example,

$$y_{i3} - \bar{y}_i$$

depends on all four observations for unit i , while

$$\Delta y_{i3} = y_{i3} - y_{i2}$$

depends only on periods 2 and 3. The two filters therefore react differently to trends, persistence, and measurement noise.

Theorem 6 (Differenced errors are mechanically serially correlated)

Suppose the idiosyncratic errors u_{it} are serially uncorrelated with variance σ_u^2 . Then the differenced errors satisfy

$$\text{Var}(\Delta u_{it}) = 2\sigma_u^2, \quad (20)$$

$$\text{Cov}(\Delta u_{it}, \Delta u_{i,t-1}) = -\sigma_u^2, \quad (21)$$

$$\text{Cov}(\Delta u_{it}, \Delta u_{is}) = 0 \quad \text{if } |t - s| > 1. \quad (22)$$

Proof 6

Because $\Delta u_{it} = u_{it} - u_{i,t-1}$,

$$\begin{aligned} \text{Var}(\Delta u_{it}) &= \text{Var}(u_{it}) + \text{Var}(u_{i,t-1}) - 2\text{Cov}(u_{it}, u_{i,t-1}) \\ &= \sigma_u^2 + \sigma_u^2 - 0 = 2\sigma_u^2. \end{aligned}$$

For adjacent differenced errors,

$$\begin{aligned} \text{Cov}(\Delta u_{it}, \Delta u_{i,t-1}) &= \text{Cov}(u_{it} - u_{i,t-1}, u_{i,t-1} - u_{i,t-2}) \\ &= \text{Cov}(u_{it}, u_{i,t-1}) - \text{Cov}(u_{it}, u_{i,t-2}) \\ &\quad - \text{Var}(u_{i,t-1}) + \text{Cov}(u_{i,t-1}, u_{i,t-2}) \\ &= -\sigma_u^2, \end{aligned}$$

using serial uncorrelatedness. If $|t - s| > 1$, the two differences share no common u term, so every covariance term is zero.

This theorem explains an important practical point. Even when the original disturbances are serially independent, the differenced disturbances are not. Adjacent differenced residuals are negatively correlated because $u_{i,t-1}$ enters once with a minus sign and once with a plus sign. That is one reason why inference in first-difference regressions should still be clustered at the unit level.

Efficiency intuition. There is no universal ranking of fixed effects and first differences. The comparison depends on the error process.

- If u_{it} is roughly serially uncorrelated, fixed effects often uses the information more efficiently because it averages against the full unit mean rather than only one neighboring period.
- If u_{it} has a strong random-walk or highly persistent component, differencing can be appealing because nearby comparisons remove that low-frequency noise more aggressively.

In short, fixed effects is often a good default in static panels, but first differences can be attractive when the economics suggests that change-from-last-period is the natural object.

Measurement error and why differencing can hurt. Suppose the observed regressor is measured with classical error:

$$x_{it}^{obs} = x_{it}^* + v_{it}, \quad \mathbb{E}[v_{it} \mid x_{it}^*, \alpha_i, u_{it}] = 0.$$

Then

$$\Delta x_{it}^{obs} = \Delta x_{it}^* + (v_{it} - v_{i,t-1}). \quad (23)$$

Equation (23) is the key warning. If v_{it} is transitory, the noise term in differences contains *two* measurement-error draws. So the signal-to-noise ratio can deteriorate sharply after differencing. Fixed effects also suffers from measurement error, but first differencing is often more fragile when the observed regressor is already noisy.

Common mistake. Students sometimes think first differencing is automatically “safer” because it looks more local. But locality is not the same as efficiency. If the regressor is noisy, differencing may turn a manageable measurement-error problem into a severe attenuation-bias problem.

Trends and interpretation. First differences are especially natural when the substantive question is itself about changes. For example:

- How does *growth* in income respond to *growth* in credit?
- How does the *change* in pollution respond to a new environmental regulation?
- How does the year-to-year *change* in employment respond to a tax change?

In such cases, the differenced equation is not just an econometric device; it can also match the economics of the problem.

A balanced way to compare the two estimators. For introductory work, the safest summary is:

1. Both FE and FD eliminate time-invariant unit heterogeneity.
2. FE uses deviations from the unit mean; FD uses period-to-period changes.
3. FE and FD are identical when $T = 2$.
4. For $T > 2$, neither estimator dominates in all environments.
5. The choice should be guided by the likely serial correlation of u_{it} , the quality of measurement of x_{it} , and the economic meaning of “within-unit variation” in the application.

Why this matters. Fixed effects and first differences are alternative filters, not universal ranking devices. The sensible choice depends on what kind of within-unit comparison is economically persuasive and on how the noise behaves in the data.

Two-Way Fixed Effects

Unit effects solve one problem: permanent differences across units. But many empirical settings face a second problem at the same time. Some shocks are common across all units in a given period. Inflation, national legislation, macroeconomic booms and recessions, commodity-price cycles, and countrywide changes in financial conditions can all move the outcome and the regressors together. If those common shocks are omitted, one-way fixed effects are not enough.

The two-way specification. That motivates the two-way fixed-effects model:

$$y_{it} = x'_{it}\beta + \alpha_i + \lambda_t + u_{it}. \quad (24)$$

Here

- α_i absorbs time-invariant differences across units;
- λ_t absorbs shocks common to all units in period t ;
- u_{it} contains the remaining unit-time-specific disturbance.

The identifying variation is therefore narrower than in the one-way fixed-effects model. We no longer compare a unit merely to its own long-run mean. We compare the unit to its own mean *and* net out the period-specific mean movement that is shared by everyone else.

Why time effects matter economically. Suppose we regress firm investment on cash flow. In a boom year, many firms simultaneously experience higher sales, higher cash flow, and higher investment. One-way fixed effects remove permanent differences across firms, but they do not remove the boom itself. Without time effects, the regression can mistake a common macro shock for a firm-level cash-flow effect. The role of λ_t is to block exactly that channel.

Why this matters. One-way fixed effects answer: how does a unit differ from its own average? Two-way fixed effects answer a stricter question: how does a unit differ from its own average *after removing what was common to everyone in that period*? That is often the relevant comparison in macro-sensitive panels.

The double-demeaning transformation. For a balanced panel, define

$$\bar{y}_i = \frac{1}{T} \sum_{t=1}^T y_{it}, \quad \bar{y}_t = \frac{1}{N} \sum_{i=1}^N y_{it}, \quad \bar{y} = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T y_{it},$$

and analogously for x_{it} and u_{it} . The two-way within transformation is

$$\tilde{y}_{it} = y_{it} - \bar{y}_i - \bar{y}_t + \bar{y}, \quad \tilde{x}_{it} = x_{it} - \bar{x}_i - \bar{x}_t + \bar{x}. \quad (25)$$

This is often called *double demeaning*. The first subtraction removes the unit mean, the second removes the time mean, and adding back the grand mean avoids subtracting the overall level twice.

Theorem 7 (Double demeaning removes additive unit and time effects)

Consider the balanced-panel model (24). Then the transformed variables in (25) satisfy

$$\tilde{y}_{it} = \tilde{x}_{it}'\beta + \tilde{u}_{it}, \quad (26)$$

where

$$\tilde{u}_{it} = u_{it} - \bar{u}_i - \bar{u}_t + \bar{u}.$$

In particular, the additive components α_i and λ_t disappear from the transformed regression.

Proof 7

Take the unit mean of (24):

$$\bar{y}_i = \bar{x}_i'\beta + \alpha_i + \bar{\lambda} + \bar{u}_i,$$

where $\bar{\lambda} = T^{-1} \sum_{t=1}^T \lambda_t$. Take the time mean:

$$\bar{y}_t = \bar{x}_t'\beta + \bar{\alpha} + \lambda_t + \bar{u}_t,$$

where $\bar{\alpha} = N^{-1} \sum_{i=1}^N \alpha_i$. Finally, take the grand mean:

$$\bar{y} = \bar{x}'\beta + \bar{\alpha} + \bar{\lambda} + \bar{u}.$$

Now subtract the unit mean and time mean from (24) and add back the grand mean:

$$\begin{aligned} y_{it} - \bar{y}_i - \bar{y}_t + \bar{y} &= (x_{it} - \bar{x}_i - \bar{x}_t + \bar{x})' \beta \\ &\quad + (\alpha_i - \bar{\alpha}_i - \bar{\alpha} + \bar{\alpha}) \\ &\quad + (\lambda_t - \bar{\lambda}_t - \bar{\lambda} + \bar{\lambda}) \\ &\quad + (u_{it} - \bar{u}_i - \bar{u}_t + \bar{u}). \end{aligned}$$

The α_i and λ_t terms cancel exactly, yielding (26).

The theorem gives the algebra, but the interpretation is the real payoff. A regressor matters in a two-way fixed-effects regression only to the extent that it varies *within unit over time* and also differs from the aggregate movement in that same period.

What variation is left after two-way fixed effects? It helps to read (25) verbally. The transformed regressor

$$\tilde{x}_{it} = x_{it} - \bar{x}_i - \bar{x}_t + \bar{x}$$

is the part of x_{it} that is:

1. not explained by the unit's average level;
2. not explained by the period's average level;
3. centered back around the overall grand mean.

So if every unit experiences the same increase in x in a given year, that common increase is absorbed by λ_t and contributes nothing to identification.

Theorem 8 (Regressors with only unit or only time variation are not identified)

Under two-way fixed effects, any regressor of the form

$$x_{it} = a_i + b_t$$

has transformed value $\tilde{x}_{it} = 0$ for all (i, t) . Consequently, its slope coefficient is not separately identified in the two-way fixed-effects regression.

Proof 8

If $x_{it} = a_i + b_t$, then

$$\bar{x}_i = a_i + \bar{b}, \quad \bar{x}_t = \bar{a} + b_t, \quad \bar{x} = \bar{a} + \bar{b}.$$

Substituting into (25),

$$\begin{aligned} \tilde{x}_{it} &= (a_i + b_t) - (a_i + \bar{b}) - (\bar{a} + b_t) + (\bar{a} + \bar{b}) \\ &= 0. \end{aligned}$$

Therefore such a regressor is perfectly collinear with the additive unit and time effects and cannot be estimated separately.

This theorem bundles together several facts that students often memorize separately:

- time-invariant regressors, such as gender or soil quality, are absorbed by unit fixed effects;
- purely aggregate regressors, such as a national interest rate that is identical for all units in a given year, are absorbed by time effects;
- only regressors with *idiosyncratic within-unit and within-time-cell variation* survive both sweeps.

Example 2

Suppose x_{it} is the statutory national minimum wage and the sample consists of regions within a single country. In a given year, the national minimum wage is the same for every region. Once year dummies are included, this regressor has no cross-regional variation left within that year, so its direct coefficient is not identified in a standard regional two-way fixed-effects regression. To estimate an effect, one would need differential exposure across regions, for example an interaction between the national policy and each region's pre-existing share of low-wage workers.

A useful way to visualize two-way fixed effects. Imagine a table with units in rows and years in columns. One-way fixed effects subtract the row means. Time effects subtract the column means. Two-way fixed effects keep only the part of each cell that remains after both sweeps. In that sense, the estimator works with deviations from the row pattern and from the column pattern at the same time.

Orthogonality condition. The natural strict-exogeneity condition for two-way fixed effects is

$$\mathbb{E}[u_{it} \mid \alpha_i, \lambda_1, \dots, \lambda_T, x_{i1}, \dots, x_{iT}] = 0 \quad \text{for each } t. \quad (27)$$

Under (27), the transformed regressor $\tilde{x}_{it}$ is orthogonal to the transformed error $\tilde{u}_{it}$. The coefficient β is then identified from the residualized variation left after removing both kinds of additive heterogeneity.

What two-way fixed effects do not solve. Two-way fixed effects are powerful, but the scope of the correction is exact. They remove:

- time-invariant unit heterogeneity, through α_i ;
- period-specific common shocks, through λ_t .

They do *not* remove:

- unit-specific shocks that vary over time,

- omitted variables that evolve differently across units,
- feedback from outcomes or shocks to future regressors,
- treatment-effect heterogeneity problems in staggered-adoption designs.

So two-way fixed effects is best viewed as a sharper additive-control strategy, not as a universal causal design.

Common mistake. Unit fixed effects do not absorb aggregate time shocks, and time fixed effects do not absorb unit-specific time-varying confounders. Two-way fixed effects removes only the additive combination of permanent unit heterogeneity and common period shocks.

Why this section matters for later lecture topics. Two-way fixed effects is not just another regression specification. It is the backbone of many empirical designs students will meet later, especially difference-in-differences and event studies. Understanding exactly what is absorbed, what variation remains, and what assumptions are still needed is therefore essential before moving to modern causal panel methods.

Inference and Clustering

Estimation asks which variation identifies β . Inference asks how variable $\hat{\beta}$ is across repeated samples. In panel data those are distinct problems because rows from the same unit are rarely independent over time. After pooling, demeaning, differencing, or adding time effects, the OLS variance still has the generic form

$$\text{Var}(\hat{\beta} \mid X^*) = (X^{*'} X^*)^{-1} X^{*'} \Omega X^* (X^{*'} X^*)^{-1}, \quad (28)$$

where Ω is the covariance matrix of the transformed disturbances. The entire inference problem is therefore a question about the structure of Ω .

If units are independent of one another but errors may be arbitrarily correlated within unit, then the natural variance estimator is cluster-robust. Writing the transformed regression for unit i as

$$y_i^* = X_i^* \beta + u_i^*,$$

the cluster-robust variance estimator is

$$\widehat{\text{Var}}(\hat{\beta}) = (X^{*'} X^*)^{-1} \left(\sum_{i=1}^N X_i^{*'} \hat{u}_i^* \hat{u}_i^{*'} X_i^* \right) (X^{*'} X^*)^{-1}. \quad (29)$$

Relative to ordinary heteroskedasticity-robust errors, the only conceptual difference is crucial: clustering keeps the within-unit residual cross-products $\hat{u}_{it}\hat{u}_{is}$ for $t \neq s$ instead of forcing them to zero.

Why are those cross-products so important? Because panel transformations themselves create dependence. We already saw this for first differences. The same logic holds for the within transformation.

Theorem 9 (Demeaned errors are mechanically correlated)

Suppose u_{it} is serially uncorrelated with mean zero and variance σ_u^2 , and define

$$\ddot{u}_{it} = u_{it} - \bar{u}_i, \quad \bar{u}_i = \frac{1}{T} \sum_{t=1}^T u_{it}.$$

Then for $t \neq s$,

$$\text{Var}(\ddot{u}_{it}) = \sigma_u^2 \left(1 - \frac{1}{T}\right), \quad (30)$$

$$\text{Cov}(\ddot{u}_{it}, \ddot{u}_{is}) = -\frac{\sigma_u^2}{T}. \quad (31)$$

Proof 9

Because the original disturbances are serially uncorrelated,

$$\text{Var}(\bar{u}_i) = \text{Var}\left(\frac{1}{T} \sum_{r=1}^T u_{ir}\right) = \frac{\sigma_u^2}{T} \quad \text{and} \quad \text{Cov}(u_{it}, \bar{u}_i) = \frac{\sigma_u^2}{T}.$$

Therefore

$$\text{Var}(\ddot{u}_{it}) = \text{Var}(u_{it}) + \text{Var}(\bar{u}_i) - 2 \text{Cov}(u_{it}, \bar{u}_i) = \sigma_u^2 \left(1 - \frac{1}{T}\right).$$

For $t \neq s$,

$$\begin{aligned} \text{Cov}(\ddot{u}_{it}, \ddot{u}_{is}) &= \text{Cov}(u_{it} - \bar{u}_i, u_{is} - \bar{u}_i) \\ &= \text{Cov}(u_{it}, u_{is}) - \text{Cov}(u_{it}, \bar{u}_i) - \text{Cov}(u_{is}, \bar{u}_i) + \text{Var}(\bar{u}_i) \\ &= 0 - \frac{\sigma_u^2}{T} - \frac{\sigma_u^2}{T} + \frac{\sigma_u^2}{T} = -\frac{\sigma_u^2}{T}. \end{aligned}$$

So even if the original shocks are independent over time, the within-transformed shocks are not. Fixed effects remove the mean component α_i , but they do not generally deliver independent residuals. That is the algebraic reason clustering is typically the default in panel applications.

Which level should be clustered? Cluster along the dimension at which dependence plausibly remains. In a worker-year panel, that is often the worker. In a firm-quarter panel, often the firm. If shocks also

comove across units within a given year, one may need time clustering or two-way clustering. A useful applied rule is that if treatment varies only at a coarse level, inference should usually respect that level. If a state policy is assigned at the state-year level, county-year observations inside the same state are not independent sources of policy variation.

Few clusters. Cluster asymptotics rely on the number of clusters, not on the number of rows. A dataset with 40,000 observations but only 18 state clusters is still primarily an 18-cluster inference problem. With few clusters, conventional clustered t statistics can over-reject. That is why good empirical work reports the number of clusters, uses cluster-based degrees-of-freedom adjustments, and in very small-cluster settings often turns to procedures such as the wild cluster bootstrap.

Common mistake. Reporting heteroskedasticity-robust standard errors without clustering in a panel is usually not a harmless shortcut. It often treats dependent observations as if they were independent and therefore overstates precision.

Practical Diagnostics

Panel econometrics is not only about formulas. It is also about learning to diagnose where identification comes from and how much independent information supports it.

Start with the source of variation. Ask whether the coefficient is identified from between-unit differences, within-unit changes, or within-unit changes net of common time effects. If a regressor barely moves within unit, then a fixed-effects estimate may be imprecise even in a very large panel.

Separate time effects from clustering. Time fixed effects and clustered standard errors do different jobs. Time effects remove average shocks common to all units in a period. Clustering changes the variance estimate to allow remaining dependence. One is about bias from omitted aggregate shocks; the other is about correct uncertainty quantification. They are complements, not substitutes.

Compare specifications economically, not mechanically. Comparing pooled OLS, FE, FD, and FE with time effects is informative because each estimator uses different variation. Large coefficient movements often reveal that between and within relationships are not the same,

or that common time shocks are important. That is substantive information, not just a robustness ritual.

Report the inferential design clearly. An empirical table should tell the reader which fixed effects are included, along which dimension standard errors are clustered, and how many clusters there are. “Robust standard errors” is too vague in panel work.

Use a short checklist. Before trusting a panel estimate, ask:

1. What variation identifies the coefficient?
2. Which observations may share shocks?
3. How many independent clusters are there?
4. Does the regressor move enough within units?
5. Do the conclusions survive reasonable alternative fixed-effects and clustering choices?

Why this matters. If a coefficient is significant with heteroskedasticity-robust errors but not with clustered errors, the right conclusion is not that clustering “changed the result.” The right conclusion is that the unclustered calculation was acting as if the panel contained more independent information than it really did.

Lecture 1 R Companions

Lecture 1 is paired with small companion scripts in `lectures/code/panel-data/`. They are designed to isolate one idea at a time rather than bury the logic inside one long file.

- `01_within_between_variation.R`: decomposes panel variation into overall, between-unit, and within-unit pieces.
- `02_pooled_ols_vs_fixed_effects.R`: shows why pooled OLS is biased when regressors are correlated with unit effects and verifies the equivalence of least-squares dummy variables and within estimation.
- `03_first_differences_vs_fixed_effects.R`: shows the exact equality of FE and FD when $T = 2$ and then illustrates why FD can be more fragile under measurement error.
- `04_two_way_fixed_effects.R`: shows how common time shocks can bias one-way FE and why regressors with only aggregate time variation disappear under two-way fixed effects.

- `05_clustered_inference.R`: compares naive, heteroskedasticity-robust, and cluster-robust uncertainty in a panel with within-unit dependence.

All scripts source `panel_helpers.R`, use only base R, and print short interpretations so that students can connect the output directly to the lecture arguments.

Lecture 1 Summary

Lecture 1 establishes a clean hierarchy of static panel methods and how to use them responsibly:

- Pooled OLS is simple but generally mixes within and between variation.
- Fixed effects remove time-invariant unit heterogeneity by demeaning.
- First differences remove the same heterogeneity by comparing adjacent periods.
- Two-way fixed effects additionally remove common time shocks.
- Credible inference usually requires clustered standard errors.

The unifying principle is straightforward: panel methods aim to isolate variation that remains credible after accounting for persistent heterogeneity, and then to measure uncertainty using the number of genuinely independent pieces of information rather than the raw number of rows.

Lecture 2: Modern Methods and Extensions

Difference-in-Differences as a Panel Estimator

Difference-in-differences (difference-in-differences (DiD)) begins with a comparison problem. Suppose a labor-market reform is introduced in some regions but not others, and we want to know whether the reform changed the employment rate. For treated regions after the reform, we observe what actually happened. We do *not* observe what would have happened in those same regions had the reform never been adopted. That missing untreated path is the counterfactual object that the design must recover.

The first temptation is a before-after comparison for the treated group:

$$\mathbb{E}[y_{post} \mid D = 1] - \mathbb{E}[y_{pre} \mid D = 1]. \quad (32)$$

This fails whenever aggregate conditions also change between the two periods. A recession, demographic shift, or commodity-price shock will appear in (32) whether or not the policy mattered.

The second temptation is a treated-control comparison in the post period:

$$\mathbb{E}[y_{post} \mid D = 1] - \mathbb{E}[y_{post} \mid D = 0]. \quad (33)$$

This fails whenever treated and control units differ systematically even before treatment. A rich region and a poor region may have different employment levels for many persistent reasons that have nothing to do with the policy under study.

Common mistake. DiD is not “treatment minus control after the policy.” That post-period gap still mixes treatment effects with permanent differences across groups.

The canonical DiD design combines both differences. Let $D_i = 1$ for treated units and $Post_t = 1$ in the post-treatment period. The sample estimator is

$$\hat{\delta}^{DiD} = (\bar{y}_{1,post} - \bar{y}_{1,pre}) - (\bar{y}_{0,post} - \bar{y}_{0,pre}), \quad (34)$$

where the first subscript indexes treatment status and the second indexes time. This is the “change in the treated group” minus the “change in the control group.”

	Pre	Post
Control ($D_i = 0$)	$\bar{y}_{0,pre}$	$\bar{y}_{0,post}$
Treated ($D_i = 1$)	$\bar{y}_{1,pre}$	$\bar{y}_{1,post}$

Table 1: The four cell means in the canonical 2x2 DiD design.

The logic is plain once the four means are laid out:

- the treated-control difference in the pre period, $\bar{y}_{1,pre} - \bar{y}_{0,pre}$, captures baseline level differences;
- the control-group change, $\bar{y}_{0,post} - \bar{y}_{0,pre}$, captures common time shocks affecting both groups;
- subtracting both leaves the treated group’s excess change after treatment.

The regression representation of the 2x2 design is

$$y_{it} = \alpha + \gamma D_i + \lambda Post_t + \delta(D_i \times Post_t) + u_{it}. \quad (35)$$

Here α is the control-group mean in the pre period, γ is the permanent treated-control gap, λ is the common post-period shock, and δ is the DiD contrast. A full proof that the interaction coefficient in (35) equals (34) is placed in [Appendix A1](#).

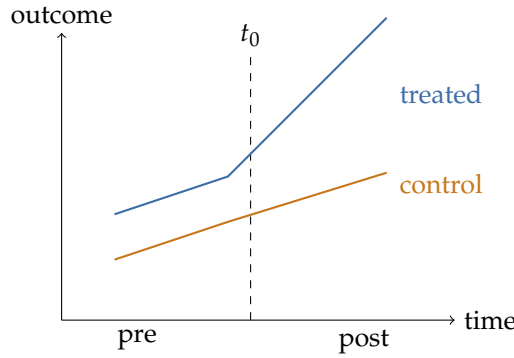


Figure 1: The untreated change for controls is used to proxy for the missing untreated change of treated units.

Why this matters. Panel structure helps because it lets us compare a unit to its own past. But that alone does not identify a causal effect. DiD works only when the untreated units supply a credible estimate of how treated units would have evolved without treatment.

A stylized worked example. Keep in mind a simple policy design. Let i index regions and let y_{it} be the regional employment rate. A labor-market reform is adopted in a subset of regions after year t_0 . The treatment group consists of reform regions, the control group of otherwise similar regions that do not adopt in that period. We will return to this example repeatedly because it forces us to ask the right empirical question: *would treated regions have followed the same untreated trend as the comparison regions absent the reform?*

Potential Outcomes and the DiD Estimand

The algebra of DiD is simple, but causal interpretation requires more discipline. Let $Y_{it}(0)$ denote the untreated potential outcome for unit i at time t , and let $Y_{it}(1)$ denote the treated potential outcome. In the canonical two-period design, treatment arrives only in the post period for treated units.

Definition 3

The average treatment effect on the treated in the post period is

$$ATT = E[Y_{i,post}(1) - Y_{i,post}(0) \mid D_i = 1]. \quad (36)$$

This definition is the correct causal target. It asks: among the units that are actually treated, how much higher or lower is the post-treatment outcome relative to the untreated outcome those same units would have experienced at the same date?

The problem is immediate. For treated units in the post period, we observe $Y_{i,post}(1)$ but not $Y_{i,post}(0)$. So the difficult object is not the

treated outcome. It is the missing untreated potential outcome for the treated group after treatment.

Observed outcomes can be written as

$$Y_{i,pre} = Y_{i,pre}(0), \quad Y_{i,post} = D_i Y_{i,post}(1) + (1 - D_i) Y_{i,post}(0).$$

Therefore the expected DiD contrast satisfies

$$\begin{aligned} \mathbb{E}[\hat{\delta}^{DiD}] &= \mathbb{E}[Y_{i,post} \mid D_i = 1] - \mathbb{E}[Y_{i,pre} \mid D_i = 1] \\ &\quad - \left(\mathbb{E}[Y_{i,post} \mid D_i = 0] - \mathbb{E}[Y_{i,pre} \mid D_i = 0] \right) \\ &= ATT \\ &\quad + \left(\mathbb{E}[Y_{i,post}(0) - Y_{i,pre}(0) \mid D_i = 1] \right. \\ &\quad \left. - \mathbb{E}[Y_{i,post}(0) - Y_{i,pre}(0) \mid D_i = 0] \right). \end{aligned} \quad (37)$$

Equation (37) is the heart of the design. The DiD contrast equals the causal effect of interest plus a term measuring how differently treated and control units would have evolved even without treatment.

The full derivation is kept in [Appendix A2](#) so the main text can stay focused on the economic idea rather than the bookkeeping.

Why this matters. Potential outcomes clarify what DiD is trying to learn. The design is not estimating a vague “policy coefficient.” It is trying to recover the missing untreated post-period outcome for the treated group.

Parallel Trends and Causal Interpretation

The central identifying assumption is now easy to state.

Assumption 1 (Parallel trends in untreated potential outcomes)

$$\mathbb{E}[Y_{i,post}(0) - Y_{i,pre}(0) \mid D_i = 1] = \mathbb{E}[Y_{i,post}(0) - Y_{i,pre}(0) \mid D_i = 0]. \quad (38)$$

Under (38), the second term in (37) is zero, so the DiD contrast identifies the *ATT*:

$$\mathbb{E}[\hat{\delta}^{DiD}] = ATT.$$

This assumption deserves very careful wording.

- It is a statement about *untreated* potential outcomes.
- It is a statement about *changes*, not levels.
- It allows treated and control groups to differ permanently in baseline means.

- It does *not* require the observed post-treatment outcomes to evolve in parallel after treatment.

That last point is often the one students miss. After treatment, observed outcomes should differ if the treatment has an effect. Parallel trends concerns the path of $Y_{it}(0)$, not the path of the observed Y_{it} once one group has been exposed to the intervention.

Common mistake. Equal baseline levels are not the identifying assumption. Treated and control groups can start far apart. What matters is whether, in the absence of treatment, their average untreated *changes* would have been the same.

Why equal levels are unnecessary. Suppose treated regions are structurally more urban than control regions. Then employment rates may differ by level for all dates because of industrial composition, education, or labor-force attachment. DiD can still be informative if those permanent level differences are stable and if untreated outcome growth would have been similar across groups.

What makes parallel trends plausible? Usually not a regression output. The case is strongest when treated and control units face the same macro environment, similar underlying demand conditions, and comparable institutional exposure, with treatment arriving for reasons that are not themselves responses to idiosyncratic outcome shocks.

Conditional parallel trends. Sometimes a more plausible claim is

$$\mathbb{E}[Y_{i,\text{post}}(0) - Y_{i,\text{pre}}(0) \mid D_i = 1, X_i] = \mathbb{E}[Y_{i,\text{post}}(0) - Y_{i,\text{pre}}(0) \mid D_i = 0, X_i]$$

for a set of pre-treatment covariates X_i . This can be useful, but it should be interpreted correctly. Controls may help align treated and control groups on observables or improve precision. They do not rescue a design in which untreated trends are plainly different for deeper economic reasons.

Threats to Identification

Once the identifying assumption is explicit, the usual threats become easier to organize. Each one is simply a reason why (38) might fail.

Differential pretrends. This is the classic failure mode. If treated regions were already declining relative to controls before the reform, a negative post-treatment DiD estimate could simply continue an existing divergence.

Threat	Why it is dangerous	What helps assess credibility
Differential pre-trends	Treated units were already on a different untreated path before the policy.	Plot long pre-period trends, compare industry mix, and explain policy timing institutionally.
Anticipation	Behavior changes before the official treatment date, contaminating the pre period.	Use institutional timelines, placebo treatment dates, and event-study leads.
Composition change	Entry, exit, migration, or sample attrition changes who is being compared over time.	Track sample stability and report balanced-sample checks where relevant.
Spillovers	Control units are indirectly affected by treatment, so they are not untreated counterfactuals.	Use geographically or economically insulated controls and discuss exposure channels.
Concurrent shocks	Another policy or shock hits treated units at the same time as the intervention.	Build the timeline carefully and control for or discuss concurrent reforms.
Endogenous timing	Units adopt treatment in response to shocks already affecting outcomes.	Explain assignment rules and test for pre-adoption outcome movements.

Table 2: Main threats to identification in DiD.

Anticipation. If firms, workers, or households adjust behavior once the policy is announced but before it legally takes effect, the measured pre period is no longer untreated. In that case the benchmark period is contaminated.

Composition change. Suppose the reform induces migration, firm entry, or selective survey response. Then the comparison over time is no longer about the same underlying population. DiD has not only a trend problem but a sample-definition problem.

Spillovers and general equilibrium effects. If untreated regions lose workers to treated regions, if nearby firms reprice in response to the treated market, or if treated hospitals divert patients to untreated hospitals, then the control group is itself indirectly treated. This violates the idea that the control trend measures $Y_{it}(0)$.

Concurrent shocks and policy bundling. Many reforms arrive as packages. A change in school funding may coincide with curriculum reform; a tax change may coincide with enforcement reform. Unless those components are separated, DiD identifies the joint package, not the single component one might want to discuss rhetorically.

Why this matters. DiD is a design argument, not a regression ritual. The regression is only a compact way of writing the comparison. The real work lies in defending the counterfactual trend.

Difference-in-Differences as a Fixed-Effects Regression

In panel notation, the basic DiD design is a special case of two-way fixed effects:

$$y_{it} = \alpha_i + \lambda_t + \delta W_{it} + u_{it}, \quad (39)$$

where $W_{it} = 1$ for treated units in treated periods and $W_{it} = 0$ otherwise. Equation (39) says exactly what the design is using:

- unit effects α_i absorb permanent differences across units;
- time effects λ_t absorb shocks common to all units in a given period;
- the coefficient on W_{it} is identified by relative within-unit changes over time.

In the canonical two-period case, first-differencing (39) yields

$$\Delta y_i = (\lambda_{post} - \lambda_{pre}) + \delta D_i + \Delta u_i, \quad (40)$$

because $\Delta W_i = D_i$. This shows the tight connection between DiD, first differences, and fixed effects. With two periods, a first-difference regression of the outcome change on the treatment-group indicator reproduces the same design logic as the interaction model in (35).

Once covariates are added, a useful specification is

$$y_{it} = \alpha_i + \lambda_t + x'_{it}\beta + \delta W_{it} + u_{it}. \quad (41)$$

The role of x_{it} should be described with discipline. Good controls can improve precision or absorb clearly measured concurrent shocks. But they do not create identification if the untreated trend for treated units remains poorly approximated by the control group.

Common mistake. Two-way fixed effects do not manufacture causal interpretation. Unit and time effects remove additive

heterogeneity and common shocks. They do not guarantee that the comparison group supplies the right counterfactual trend.

Inference and Clustering in DiD

Inference in DiD deserves separate attention because the usual standard-error defaults are often misleading. The problem is not exotic. In panels, disturbances are rarely independent across time within a unit, and treatment is often assigned at the group level rather than at the level of each row.

Serial correlation. If u_{it} is persistent within a unit, then the number of statistically independent pieces of information is much smaller than the number of panel rows. In policy panels this matters greatly because treatment typically changes only once or a few times, while outcomes remain autocorrelated. Ignoring that persistence tends to make standard errors too small.

Clustering. As a practical rule, cluster at the level at which treatment varies and shocks are likely to be correlated. If a state law changes treatment status, clustering by individual is too fine because the policy shock is common within state. State-level clustering is usually the more relevant benchmark.

Few clusters. When the number of treated or untreated clusters is small, even cluster-robust standard errors can be unreliable. That is a warning to be explicit rather than casual. In such settings researchers often rely on additional finite-sample adjustments or resampling procedures, but the first obligation is simply to report the cluster structure clearly.

Why this matters. Time effects and clustered inference solve different problems. Time effects control for common shocks in the conditional mean. Clustering adjusts uncertainty for dependence that remains in the error process. One does not substitute for the other.

A short checklist. Before trusting a DiD estimate, ask:

1. At what level is treatment assigned?
2. At what level are residual shocks likely to be correlated?
3. How many independent clusters are there?

4. Would the headline result survive a more conservative clustering choice?

Event Studies and Dynamic Treatment Effects

Many policies do not have an immediate constant effect. Firms may adjust gradually, workers may learn about the reform over time, and administrative systems may phase in the new rules. Event studies extend DiD by allowing the treatment effect to vary with time since adoption.

Let G_i denote the treatment date for unit i . A standard specification is

$$y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k 1\{t - G_i = k\} + u_{it}, \quad (42)$$

where the omitted category $k = -1$ is the period just before treatment. The coefficients have a natural interpretation:

- for $k < 0$, β_k are *lead* coefficients describing differences before treatment relative to the reference period;
- for $k \geq 0$, β_k are *lag* coefficients describing the dynamic response after treatment.

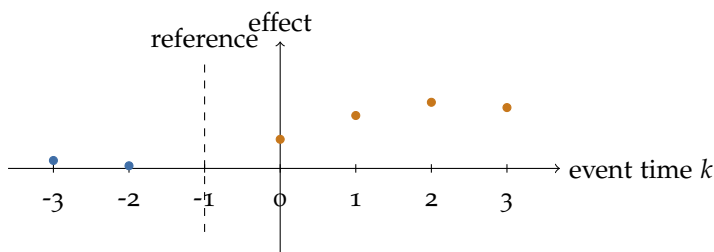


Figure 2: Event studies separate pre-treatment leads from post-treatment lags around an omitted reference period.

Event studies are useful for two different reasons. First, they provide a visual and statistical diagnostic for pre-treatment divergence. Second, they describe the substantive dynamics of treatment effects after adoption.

How to read an event-study plot without fooling yourself.

1. Check the omitted reference period. Every coefficient is relative to that baseline.
2. Look at the scale of the vertical axis before making claims about “flat” pretrends.
3. Treat insignificant leads as weak evidence at best, not proof of validity.

4. Read post-treatment coefficients as dynamic effects only if the comparison set is credible.

Common mistake. Failing to reject zero pre-treatment leads is not the same as proving parallel trends. Small samples, noisy outcomes, and poor comparison groups can all generate uninformative pretrend tests.

Staggered Adoption and Modern TWFE Cautions

Many modern applications do not have one treated group and one treatment date. Instead, different cohorts adopt at different times. This is called *staggered adoption*. The natural temptation is to estimate (39) or (42) with two-way fixed effects and interpret the treatment coefficient as the average effect of treatment.

That interpretation can fail when treatment effects are heterogeneous across cohorts or over event time. The reason is subtle but important: once some units are already treated, they may be used as controls for units treated later. But already-treated units are not untreated counterfactuals.

A three-cohort intuition. Suppose there are early-treated, late-treated, and never-treated regions.

- Early-treated regions are untreated controls only before they adopt.
- After they adopt, they should no longer serve as controls for late-treated regions.
- Standard two-way fixed effects may nevertheless use them that way.

If treatment effects grow over time, shrink over time, or differ across cohorts, the resulting coefficient can become a difficult weighted average of comparisons, some of which carry the wrong sign or the wrong economic interpretation.

Why this matters. Two-way fixed effects are not problematic because fixed effects are bad. The issue is the comparison set. Under staggered adoption, already-treated units can contaminate the counterfactual for newly treated units.

The practical lesson is straightforward. In staggered settings, one should prefer estimators organized around cohort-time or group-time average treatment effects, using not-yet-treated or never-treated

units as comparisons whenever that design is economically sensible. Students do not need the full software taxonomy at this stage, but they should leave with the right instinct: *ask who is being compared to whom in each calendar period.*

Common mistake. Seeing a tidy two-way fixed-effects coefficient in a staggered-adoption design does not guarantee that the estimate is the average causal effect one has in mind.

Applied Examples

The point of applications is not to memorize literatures. It is to practice counterfactual reasoning.

1. *Minimum wage and employment.* Treatment is a minimum-wage increase in some states; controls are nearby or otherwise comparable states without the increase. The identifying claim is that absent the policy, employment trends would have been similar. The main threat is that treated states may already differ in labor-demand trends. Credibility improves with institutional proximity, longer pre-period evidence, and industry-specific checks.

2. *Education reform.* Treatment is a compulsory-schooling change or class-size reform introduced in some cohorts, districts, or regions. Controls are unaffected cohorts or places. The main threats are staggered rollout, other school-finance reforms, and selective migration. Credibility improves if treatment timing is tightly linked to legislation rather than local educational decline.

3. *Health policy.* Treatment is an insurance expansion, hospital payment reform, or provider incentive scheme. Controls are unaffected patients, hospitals, or regions. The main threats are coding changes, patient sorting, and simultaneous administrative reforms. Credibility improves if outcomes are defined consistently and composition is carefully tracked.

4. *Environmental regulation.* Treatment is exposure to a new regulation in highly regulated regions or sectors; controls are less-exposed areas. The main threat is that heavily exposed areas may already be in industrial decline. Credibility improves with evidence on pre-regulation trends and with outcome measures less sensitive to unrelated industrial restructuring.

5. *Tax or benefit reform.* Treatment is eligibility for a new tax credit or benefit rule; controls are ineligible but otherwise comparable groups. The main threat is anticipation, because households and firms may shift timing before the reform date. Credibility improves with a clear institutional timeline and event-study evidence showing whether behavior moved before implementation.

Difference-in-Differences Summary and Practical Takeaways

Object	Identifying variation	Key assumption	Main diagnostic
2x2 DiD	Change in treated minus change in control	Parallel untreated trends	Four-cell means and pre-period design logic
TWFE DiD	Relative within-unit changes with time effects	Comparison units trace the right counterfactual trend	Treatment timing, control-group choice, clustered inference
Event study	Dynamic relative changes by event time	Credible pre-period and valid comparison set	Pre-period coefficients, scale, and institutional timing
Staggered-adoption design	Cohort-time contrasts	Not-yet-treated or never-treated units are valid controls	Check who serves as control in each period

Table 3: A compact map of DiD designs and diagnostics.

The main lessons are compact:

1. DiD solves a comparison problem by using untreated changes to proxy for a missing counterfactual change.
2. The causal estimand is the treated group's missing untreated post-treatment outcome, not a mechanical interaction coefficient by itself.
3. Parallel trends is the key identifying assumption, and it concerns untreated potential outcomes.
4. Fixed effects make the algebra convenient, but validity still depends on design, timing, and the quality of the control group.
5. Event studies are useful for diagnosis and dynamics, but they are easy to overread.
6. Under staggered adoption, standard two-way fixed effects can compare newly treated units to already-treated units in ways that distort interpretation.

Appendix to the DiD Module

Appendix A1: Why the interaction coefficient equals the 2x2 DiD contrast.

Back to the canonical 2x2 design.

Let

$$\mu_{dp} = \mathbb{E}[y_{it} \mid D_i = d, \text{Post}_t = p], \quad d, p \in \{0, 1\}.$$

Evaluate (35) in each cell:

$$\begin{aligned} \mu_{00} &= \alpha, \\ \mu_{10} &= \alpha + \gamma, \\ \mu_{01} &= \alpha + \lambda, \\ \mu_{11} &= \alpha + \gamma + \lambda + \delta. \end{aligned}$$

Subtract the pre-period treated-control gap from the post-period treated-control gap:

$$\begin{aligned} (\mu_{11} - \mu_{01}) - (\mu_{10} - \mu_{00}) &= (\alpha + \gamma + \lambda + \delta - (\alpha + \lambda)) \\ &\quad - ((\alpha + \gamma) - \alpha) \\ &= \delta. \end{aligned}$$

Equivalently,

$$\delta = (\mu_{11} - \mu_{10}) - (\mu_{01} - \mu_{00}),$$

which is exactly the population version of (34). So the interaction coefficient is not a mysterious regression artifact; it is simply the double difference written in linear-model notation.

Appendix A2: Deriving the ATT under parallel trends. **Back to the potential-outcomes discussion.**

Start from the population DiD contrast:

$$\Delta^{DiD} = \left(\mathbb{E}[Y_{i,\text{post}} \mid D_i = 1] - \mathbb{E}[Y_{i,\text{pre}} \mid D_i = 1] \right) - \left(\mathbb{E}[Y_{i,\text{post}} \mid D_i = 0] - \mathbb{E}[Y_{i,\text{pre}} \mid D_i = 0] \right).$$

Because treated units are untreated in the pre period,

$$Y_{i,\text{pre}} = Y_{i,\text{pre}}(0).$$

Because control units are untreated in both periods,

$$Y_{i,\text{post}} = Y_{i,\text{post}}(0), \quad Y_{i,\text{pre}} = Y_{i,\text{pre}}(0) \quad \text{when } D_i = 0.$$

For treated units in the post period,

$$Y_{i,\text{post}} = Y_{i,\text{post}}(1).$$

Substitute these relations into Δ^{DiD} :

$$\begin{aligned} \Delta^{DiD} &= \left(\mathbb{E}[Y_{i,\text{post}}(1) \mid D_i = 1] - \mathbb{E}[Y_{i,\text{pre}}(0) \mid D_i = 1] \right) \\ &\quad - \left(\mathbb{E}[Y_{i,\text{post}}(0) \mid D_i = 0] - \mathbb{E}[Y_{i,\text{pre}}(0) \mid D_i = 0] \right). \end{aligned}$$

Add and subtract $\mathbb{E}[Y_{i,post}(0) \mid D_i = 1]$:

$$\begin{aligned} \Delta^{DiD} = & \left(\mathbb{E}[Y_{i,post}(1) \mid D_i = 1] - \mathbb{E}[Y_{i,post}(0) \mid D_i = 1] \right) \\ & + \left(\mathbb{E}[Y_{i,post}(0) - Y_{i,pre}(0) \mid D_i = 1] \right. \\ & \left. - \mathbb{E}[Y_{i,post}(0) - Y_{i,pre}(0) \mid D_i = 0] \right). \end{aligned}$$

The first bracket is exactly the average treatment effect on the treated in the post period:

$$ATT = \mathbb{E}[Y_{i,post}(1) - Y_{i,post}(0) \mid D_i = 1].$$

The second bracket is the difference in untreated trends across treated and control groups. Under parallel trends (38), that term is zero, and therefore

$$\Delta^{DiD} = ATT.$$

This derivation makes the role of the assumption transparent: it is the step that lets the control group's untreated change stand in for the treated group's missing untreated change.

Random Effects

Random effects keep the unobserved-effects structure

$$y_{it} = x'_{it}\beta + \alpha_i + u_{it},$$

but replace the fixed-effects logic with a stronger orthogonality assumption:

$$\mathbb{E}[\alpha_i \mid x_{i1}, \dots, x_{iT}] = 0. \quad (43)$$

Under (43), the unit effect is uncorrelated with the regressors, so both within and between variation are informative about β .

The random-effects estimator can be written as a quasi-demeaned regression:

$$y_{it} - \theta \bar{y}_i = (x_{it} - \theta \bar{x}_i)' \beta + (u_{it} - \theta \bar{u}_i), \quad (44)$$

where, in the balanced-panel case,

$$\theta = 1 - \sqrt{\frac{\sigma_u^2}{\sigma_u^2 + T\sigma_\alpha^2}}. \quad (45)$$

When $\theta = 1$, we recover fixed effects. When $\theta = 0$, we are close to pooled OLS. Random effects therefore sit between those two estimators.

The attraction of random effects is efficiency. Because it uses both within and between information, it can estimate coefficients on time-invariant regressors and can be more precise than fixed effects when the orthogonality assumption is credible.

Why this matters. Random effects do not merely offer a computational shortcut. They estimate a different model with a different identifying assumption. The efficiency gain is purchased by assuming away correlation between regressors and permanent heterogeneity.

Correlated Random Effects (Mundlak)

Correlated random effects, often associated with Mundlak, provide a useful compromise. Instead of assuming α_i is orthogonal to the regressors, write

$$\alpha_i = \bar{x}_i' \pi + c_i, \quad \mathbb{E}[c_i \mid x_{i1}, \dots, x_{iT}] = 0. \quad (46)$$

Substituting into the outcome equation gives

$$y_{it} = x_{it}' \beta + \bar{x}_i' \pi + c_i + u_{it}. \quad (47)$$

Equation (47) says that the correlation between regressors and permanent heterogeneity is captured by the unit means $\bar{x}_i$. Conditional on those means, the remaining random effect c_i is orthogonal.

This has two major virtues:

- The coefficient on the time-varying regressor, β , coincides with the fixed-effects estimand under standard conditions.
- We can still estimate coefficients on time-invariant regressors if they are added to the specification.

The Mundlak formulation also has interpretive value. It separates the *within effect* of changing x_{it} from the *between effect* associated with a unit's long-run average exposure.

Common mistake. Random effects versus fixed effects is not the only choice. In many applications, correlated random effects are the right middle ground because they respect the likely correlation between unit heterogeneity and the long-run regressor level.

Fixed Effects Versus Random Effects

Fixed effects and random effects answer different econometric questions:

- **FE:** what is the within-unit association between changes in x_{it} and changes in y_{it} after removing all time-invariant heterogeneity?

- **RE:** what is the best linear combination of within and between variation under the assumption that permanent heterogeneity is orthogonal to regressors?

The practical comparison is straightforward:

- **Assumptions.** FE allows arbitrary correlation between α_i and x_{it} ; RE does not.
- **Variation used.** FE uses only within-unit variation; RE uses within and between variation.
- **Time-invariant regressors.** FE cannot estimate them; RE can.
- **Efficiency.** If (43) is correct, RE is typically more efficient.

This tradeoff motivates Hausman-style reasoning. If RE is consistent, both FE and RE are consistent, but RE is more efficient. If RE is inconsistent because α_i and x_{it} are correlated, FE remains consistent while RE does not. A large discrepancy between the two estimators is therefore evidence against the random-effects orthogonality restriction.

Why this matters. The FE-versus-RE decision is really a decision about economic heterogeneity. Do permanently different units choose systematically different regressor values? In most microeconomic settings, that is a substantive question with an economic answer, not a purely statistical one.

Dynamic Panels

Dynamic panels introduce lagged outcomes as regressors:

$$y_{it} = \rho y_{i,t-1} + x'_{it}\beta + \alpha_i + u_{it}. \quad (48)$$

This specification is natural when outcomes are persistent, as in earnings, productivity, or investment.

However, fixed effects no longer solve the endogeneity problem by themselves. Demeaning (48) gives

$$y_{it} - \bar{y}_i = \rho(y_{i,t-1} - \bar{y}_i) + (x_{it} - \bar{x}_i)'\beta + (u_{it} - \bar{u}_i).$$

The transformed lagged dependent variable is correlated with the transformed error because $y_{i,t-1}$ contains $u_{i,t-1}$, which is part of $\bar{u}_i$.

This generates *Nickell bias*: in short panels, the fixed-effects estimator of ρ is biased even when the idiosyncratic shocks are serially uncorrelated. The bias is of order $1/T$, so it becomes less important when T is large, but it can be severe when T is small.

Common mistake. Adding unit fixed effects does not make a lagged dependent variable exogenous. In dynamic panels, fixed effects remove α_i but create a new correlation between the transformed lag and the transformed error.

Dynamic panels therefore motivate internal-instrument approaches such as Arellano-Bond and related generalized method of moments estimators.

Arellano-Bond Intuition

Arellano-Bond starts from the first-differenced version of the dynamic panel:

$$\Delta y_{it} = \rho \Delta y_{i,t-1} + \Delta x'_{it} \beta + \Delta u_{it}. \quad (49)$$

Differencing removes the unit effect, but $\Delta y_{i,t-1}$ is still endogenous because it contains $u_{i,t-1}$, which is also part of $\Delta u_{it} = u_{it} - u_{i,t-1}$.

The key insight is that deeper lags of y are valid instruments under suitable assumptions. If u_{it} is serially uncorrelated, then for $s \geq 2$,

$$\mathbb{E}[y_{i,t-s} \Delta u_{it}] = 0. \quad (50)$$

So $y_{i,t-2}$ and earlier lags can instrument $\Delta y_{i,t-1}$.

This is attractive because the panel itself supplies the instruments. We do not need an external instrument in the usual instrumental variables (IV) sense. The identifying content comes from timing restrictions.

In practice, Arellano-Bond is usually implemented as a generalized method of moments estimator that stacks many such moment conditions. Two diagnostics are central:

- **Serial-correlation tests.** First-differenced errors should typically show first-order correlation but not second-order correlation if the original u_{it} is serially uncorrelated.
- **Instrument discipline.** Too many instruments can overfit endogenous variables and weaken specification tests.

Why this matters. Arellano-Bond turns the panel's time structure into an identification device. The estimator works not because differencing is magical, but because the lag structure implies moment conditions that can be defended economically.

The broader lesson is that dynamic panels are not simply a more elaborate version of static fixed effects. Once lagged outcomes en-

ter the regression, timing assumptions become central and internal instruments become the natural extension.

Lecture 2 Summary

Lecture 2 adds a causal and dynamic layer to the panel toolkit:

- DiD is a fixed-effects design built on the parallel-trends assumption.
- Event studies trace dynamics but require care under staggered adoption.
- Random effects gain efficiency by imposing orthogonality between regressors and unit effects.
- Mundlak's correlated random effects recover fixed-effects intuition while preserving more structure.
- Dynamic panels break the static fixed-effects logic and motivate internal-instrument estimators such as Arellano-Bond.

The common theme is that richer empirical questions require stronger design assumptions. Modern panel methods are powerful because they encode those assumptions explicitly, not because the panel structure alone guarantees causal interpretation.

Lecture 3: Synthetic Control

Synthetic Control and the Comparison Problem

Synthetic control is designed for a setting that is common in applied policy work and awkward for standard panel methods: treatment affects one unit, or a very small number of units, and no single untreated unit looks like an obviously credible comparison.

Suppose one state adopts a major tobacco-control law. We observe what happens in that treated state after the law. We do not observe what would have happened in that same state had the law not been adopted. That is the familiar counterfactual problem. But now there is an additional difficulty: if no single untreated state tracks the treated state's pre-treatment path closely, a simple treated-versus-control comparison is hard to defend.

The before-after comparison

$$Y_{1,post} - Y_{1,pre} \tag{51}$$

is contaminated by aggregate shocks, long-run secular trends, and any other post-period forces unrelated to treatment. A single-control

comparison is often just as fragile, because the choice of control can become arbitrary.

Why this matters. One treated unit is not an easy special case of panel econometrics. It is often a harder case, because the entire design hinges on whether the untreated comparison is genuinely credible.

The synthetic-control idea is simple: rather than selecting one untreated unit informally, construct a weighted average of untreated donor units so that the resulting synthetic unit reproduces the treated unit's pre-treatment characteristics and pre-treatment outcome path as closely as possible.

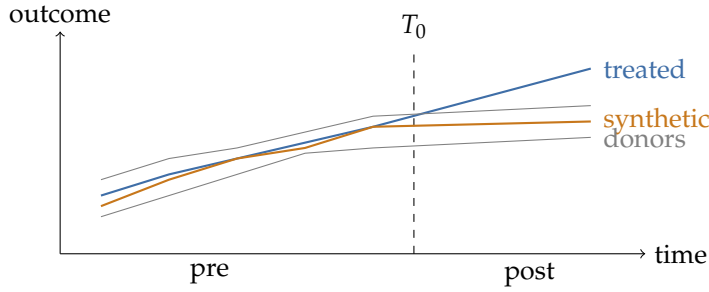


Figure 3: Synthetic control builds a weighted untreated comparison that tracks the treated unit closely before treatment.

Common mistake. A synthetic control is not the untreated unit with the closest average level. It is a weighted combination chosen to reproduce the treated unit's pre-treatment features and trajectory.

Constructing the Synthetic Control

Let unit 1 be treated, and let units $j = 2, \dots, J + 1$ be untreated donor units. Treatment begins after period T_0 , so the pre-treatment periods are $t = 1, \dots, T_0$ and the post-treatment periods are $t = T_0 + 1, \dots, T$.

The synthetic control is a weighted average of donor outcomes:

$$\hat{Y}_{1t}(0) = \sum_{j=2}^{J+1} w_j Y_{jt}, \quad t \leq T_0 \text{ and, by extrapolation, } t > T_0, \quad (52)$$

where the weights satisfy

$$w_j \geq 0, \quad \sum_{j=2}^{J+1} w_j = 1. \quad (53)$$

These restrictions matter. Nonnegative weights that sum to one ensure that the synthetic unit is a convex combination of actual untreated units. This keeps the comparison transparent and avoids unstable extrapolation based on negative weights or arbitrary linear projections.

Donor unit	Weight	Interpretation
State A	0.45	largest contributor to the synthetic control
State B	0.30	helps match medium-run pre-treatment path
State C	0.15	improves predictor balance
State D	0.10	fine-tunes pre-treatment fit

Table 4: A stylized donor-weights table for a synthetic-control design.

The immediate intuition is straightforward. If no single donor matches the treated unit well, perhaps a weighted combination will. One donor may resemble the treated unit in baseline levels, another in medium-run dynamics, and a third in observed predictors such as income, industry composition, or population structure. The synthetic unit pools those strengths.

Why this matters. The method does not ask us to believe that one untreated unit is “the control.” It asks whether a transparent weighted average of untreated units can reproduce the treated unit’s pre-treatment path well enough to serve as a credible missing untreated outcome.

Potential Outcomes and the Synthetic-Control Estimand

To keep the causal object explicit, return to potential outcomes. Let $Y_{1t}(0)$ denote the untreated potential outcome for the treated unit and $Y_{1t}(1)$ the treated potential outcome. For donor units, assume they remain untreated during the evaluation window, so we observe their untreated outcomes.

The object of interest for each post-treatment period is

$$\tau_{1t} = Y_{1t}(1) - Y_{1t}(0), \quad t > T_0. \quad (54)$$

The problem is again the same missing-data problem as in DiD: after treatment starts, we observe $Y_{1t}(1)$ but not $Y_{1t}(0)$.

Synthetic control estimates the missing untreated path with the weighted donor average, yielding

$$\hat{\tau}_{1t} = Y_{1t} - \sum_{j=2}^{J+1} \hat{w}_j Y_{jt}, \quad t > T_0. \quad (55)$$

This estimator is best understood as a constructed counterfactual path, not as a regression coefficient. The entire design lives or dies on

whether the weighted donor path is a credible proxy for $Y_{1t}(0)$ after treatment.

Why this matters. Synthetic control does not begin from a parallel-trends coefficient restriction. It begins from a reconstruction problem: can untreated donors be combined so that the treated unit's missing untreated path is well approximated?

Optimization, Weights, and Pre-Treatment Fit

The practical problem is how to choose the donor weights. Let X_1 be a vector of pre-treatment characteristics for the treated unit and let X_0 be the corresponding matrix for the donor pool. These predictors often include averages of observed covariates and, crucially, lagged values of the outcome itself.

Weights are typically chosen by solving

$$\min_W (X_1 - X_0 W)' V (X_1 - X_0 W) \quad (56)$$

subject to (53), where $W = (w_2, \dots, w_{J+1})'$ and V is a positive semidefinite weighting matrix assigning relative importance to predictor mismatch.

Equation (56) says that the method looks for donor weights that make the treated unit and the synthetic unit as similar as possible before treatment, where similarity is judged in economically chosen dimensions. If the pre-treatment outcomes are included heavily in X_1 and X_0 , then the method is explicitly trying to match the path of the treated unit before treatment.

Predictor	Treated unit	Synthetic control
Average pre-treatment outcome	121.4	120.9
Income per capita	42.0	41.8
Manufacturing share	0.19	0.20
Urbanization rate	0.73	0.71

Table 5: A stylized predictor-balance table for synthetic control.

Why pre-treatment fit matters. If the synthetic unit tracks the treated unit closely before treatment, then it becomes more plausible that the weighted donor combination is reproducing not only observed predictors but also the underlying latent forces driving the untreated outcome path. Lagged outcomes are often powerful summary statistics for those latent forces.

But fit is not everything. Good fit is evidence, not proof. It is possible to match a short pre-treatment path for accidental reasons, or to obtain decent fit while still relying on a donor pool that is economically

inappropriate. Conversely, poor pre-treatment fit is a serious warning sign. If the synthetic unit cannot reproduce the treated path before treatment, it is difficult to argue that it should recover the missing untreated path after treatment.

Common mistake. Good visual fit is not a theorem. It increases credibility, but it does not eliminate concerns about anticipation, donor contamination, or post-treatment shocks that differentially affect the treated unit for unrelated reasons.

The geometric intuition behind the convex-combination restriction is useful: the treated unit should lie in or near the convex hull of the donor units in the space of pre-treatment predictors and outcomes. If the treated unit lies far outside that hull, the method is being asked to construct a comparison that the donor pool simply cannot supply. A slightly fuller discussion is placed in [Appendix S1](#).

Synthetic Control versus Difference-in-Differences and Matching

Synthetic control belongs to the same broad family of comparative counterfactual designs as matching and DiD, but it solves a different practical problem.

Method	Comparison unit	Key identifying logic	Main diagnostic emphasis
Matching	one or a few matched units	treated and controls are similar on observables	covariate balance
DiD	untreated group over time	untreated trends for treated and control groups would have moved in parallel	pre-treatment trends and design argument
Synthetic control	weighted donor combination	a synthetic unit can reconstruct the treated unit's untreated path	pre-treatment fit and placebo gaps

Table 6: Synthetic control in relation to matching and DiD.

Relative to DiD, synthetic control is especially attractive when the number of treated units is very small and long pre-treatment histories are available. Rather than asserting that an untreated group's average trend is the right counterfactual, the method attempts to build a customized untreated comparison from donor weights.

Relative to standard regressions, the weight restrictions make the comparison more transparent. A linear regression can match the

treated unit by extrapolating outside the support of the donor units, often with negative implicit weights. Synthetic control is usually more disciplined because it keeps the synthetic unit inside the range spanned by untreated donors.

Why this matters. Synthetic control is attractive precisely when ordinary specifications feel too parametric or too coarse. It lets the data construct the comparison unit while keeping the construction visible to the reader.

Diagnostics, Placebos, and Inference

Synthetic-control evidence is not usually judged by default heteroskedasticity-robust standard errors. The number of treated units is often one, and the main empirical question is comparative: is the treated-unit post-treatment gap unusually large relative to what we would see if treatment were reassigned to untreated units?

Path plots and gap plots. The first diagnostic is always graphical. Plot the treated unit and the synthetic unit over time. Then plot the post-treatment gap

$$Gap_t = Y_{1t} - \sum_{j=2}^{J+1} \hat{w}_j Y_{jt}.$$

If the paths track each other closely before treatment and diverge only after treatment, that is the basic pattern the design hopes to uncover.

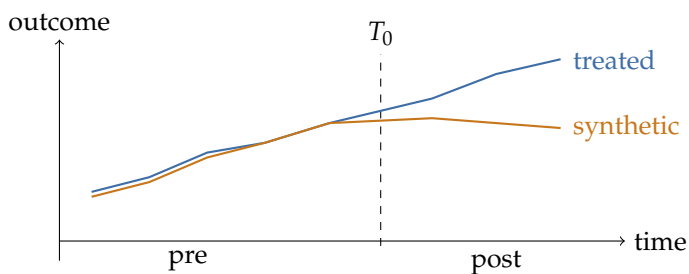


Figure 4: A canonical synthetic-control result: close pre-treatment fit and post-treatment divergence.

Placebo tests. The standard inferential device is an in-space placebo exercise. Reassign treatment to each donor unit one at a time, construct that donor's synthetic control using the remaining untreated units, and compute its post-treatment gap. This produces a distribution of placebo gaps against which the treated-unit gap can be compared.

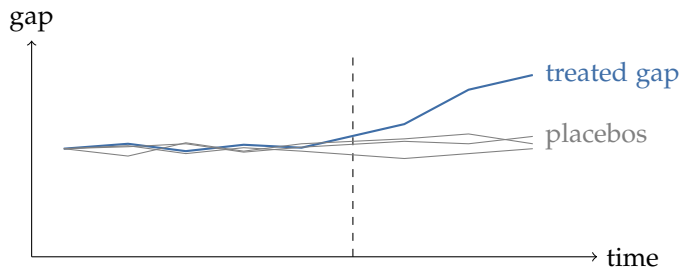


Figure 5: Placebo-gap logic: the treated-unit post-treatment gap should stand out relative to placebo gaps with similar pre-treatment fit.

Pre-treatment fit and gap scaling. Placebo exercises are only informative when the placebo units have acceptable pre-treatment fit. If a donor unit cannot itself be well approximated by a synthetic control, its placebo gap is not a useful benchmark. This is why applied work often reports pre-treatment and post-treatment root mean squared prediction error comparisons.

Sensitivity analysis. Good synthetic-control practice includes at least three additional checks:

- **Donor-pool sensitivity.** Does the result survive reasonable changes in the set of eligible donors?
- **Predictor sensitivity.** Does the result survive reasonable changes in the predictor set or in how pre-treatment outcomes are weighted?
- **Leave-one-out analysis.** Does the result disappear if the largest-weight donor is removed?

Why this matters. A large post-treatment gap means little if pre-treatment fit is poor. Synthetic-control diagnostics are not optional embellishments; they are part of the identification argument.

Common mistake. Placebo exercises are comparative diagnostics, not magic p-values. They help us judge whether the treated-unit gap is unusually large relative to plausible untreated alternatives, but they do not replace economic reasoning about the design.

Threats to Credibility and Design Limits

The method is attractive, but it is not automatic. The main threats are easy to state once the logic of the design is clear.

Threat	Why it is dangerous	What helps assess credibility
Poor pre-treatment fit	Synthetic path does not approximate the treated path even before treatment.	Longer pre-treatment horizon, balance tables, and path plots.
Donor contamination	Donor units are indirectly affected by the treatment or a related shock.	Careful donor-pool construction and institutional discussion.
Anticipation	Treated unit adjusts before the formal intervention date.	Clear policy timeline and exclusion of contaminated pre-period observations.
Outside convex hull	Treated unit cannot be reconstructed by a convex combination of donors.	Inspect weight concentration and fit quality; reconsider donor pool.
Short pre-treatment period	Matching may look good accidentally because little history is available.	Use as long a pre-treatment history as defensible.
Unrelated structural break	Post-treatment treated-unit shock is not caused by treatment.	Institutional narrative, placebo dates, and auxiliary outcomes.

Table 7: Main threats to credibility in synthetic-control designs.

Poor pre-treatment fit. This is the clearest red flag. If the synthetic unit fails before treatment, it is very hard to defend it after treatment.

Donor-pool contamination. If the donors are themselves touched by the policy, exposed through spillovers, or affected by the same reform package, then they are not untreated counterfactuals. The method cannot repair a contaminated donor pool by optimization alone.

Outside the convex hull. Sometimes the treated unit is simply too unusual. In that case most of the weight may pile onto one or two implausible donors, or fit may remain poor even after optimization. That is not a minor nuisance. It means the comparison problem may not be solvable with the available donor pool.

Short pre-treatment windows and structural breaks. With very little pre-treatment data, the method can overstate credibility because many donor combinations may fit a short path by chance. And even with excellent fit, a large post-treatment gap is not necessarily causal if a non-treatment structural break hits the treated unit at the same time.

Common mistake. Synthetic control does not rescue a bad research design. If the donor pool is economically inappropriate or if the treated unit is fundamentally incomparable, the optimization problem will still produce weights, but those weights do not create credibility.

Extensions and Modern Variants

The original synthetic-control method was developed for the case of one treated unit and a small donor pool, but the underlying idea has generated a broader family of approaches.

Multiple treated units. When several units are treated, one can aggregate treated units, build separate synthetic controls, or use methods designed for multi-treated settings.

Augmented synthetic control. One limitation of the original method is that convex-combination restrictions can produce imperfect fit when the treated unit lies near but not inside the donor convex hull. Augmented approaches combine synthetic control with outcome-model adjustment to reduce bias from residual mismatch.

Interactive fixed effects and matrix completion. These methods generalize the idea that untreated outcomes may be driven by latent common factors with unit-specific loadings. Conceptually, they sit near synthetic control because they also try to reconstruct missing untreated outcomes using the panel's low-rank structure.

Synthetic DiD. This newer bridge method combines ideas from DiD and synthetic control, weighting units and sometimes periods to improve balance before applying a difference-in-differences style comparison. The practical lesson is not software taxonomy. It is that modern causal panel methods increasingly focus on constructing better comparison objects rather than simply running standard regressions.

Applied Examples

The purpose of applications is to sharpen judgment about when synthetic control is persuasive.

1. *Tobacco control and cigarette consumption.* Treatment is a major anti-smoking law in one state. The donor pool is untreated states over the

same period. Synthetic control is attractive because no single comparison state closely tracks the treated state's smoking path before the law. The key concern is that public-health attitudes may already have been evolving differently in the treated state.

2. *German reunification and macroeconomic outcomes.* Treatment is reunification for West Germany or unified Germany, depending on the design. The donor pool is a set of comparable developed economies. Synthetic control is attractive because no single country is an obvious counterfactual. The main concern is that the historical event may also alter international spillovers and comparison-country dynamics.

3. *Terrorism, conflict, or violence shocks.* Treatment is a conflict shock to one region or country. The donor pool is untreated regions or countries. Synthetic control is attractive because such shocks are often concentrated in one location. The main concern is donor contamination through trade, migration, or regional instability.

4. *Congestion pricing or urban transport reform.* Treatment is a major city-level transport intervention. The donor pool is comparable untreated cities. The main attraction is that the treated city may be unique, making naive city-by-city comparisons weak. The main concern is that broader urban trends or complementary investments may coincide with the reform.

5. *Environmental or public-health regulation.* Treatment is a state or country regulation affecting pollution, emissions, or health behavior. The donor pool is untreated jurisdictions. The key concern is whether the treated jurisdiction was already on a distinct path because of industry composition or prior regulatory momentum.

Synthetic Control Summary and Practical Takeaways

The main lessons are compact:

1. Synthetic control is a transparent way to build a comparison unit when no single untreated unit is credible.
2. The method estimates a missing untreated path, not a regression coefficient with automatic causal meaning.
3. Pre-treatment fit is central because the method is credible only if the treated unit is well reconstructed before treatment.
4. Good fit is persuasive evidence, not definitive proof.

Object	Identifying logic	Main diagnostic	Main threat
Synthetic path	weighted donor combination approximates untreated treated-unit path	close pre-treatment fit	poor donor support
Post-treatment gap	treated minus synthetic path after T_0	divergence after treatment with stable pre-fit	unrelated structural break
Placebo comparison	treated gap is unusual relative to donor-placebo gaps	treated gap stands out among well-fit placebos	poor placebo comparability
Sensitivity analysis	result survives reasonable design changes	donor and predictor robustness	fragile dependence on one donor or one specification

Table 8: A compact map of the synthetic-control design.

5. Diagnostics, placebo exercises, and donor-pool sensitivity are part of the identification argument, not optional extras.

Appendix to the Synthetic Control Module

Appendix S1: Geometry and the convex-hull intuition. [Back to the synthetic-control construction.](#)

The convex-combination restriction in (53) means that the synthetic unit must lie inside the set of all weighted averages of donor predictors and donor pre-treatment outcomes. Geometrically, that set is the convex hull of the donor units.

This matters because it clarifies what the method can and cannot do. If the treated unit lies inside that hull, then exact or near-exact pre-treatment matching may be possible without extrapolation. If it lies outside the hull, the method can only approximate it imperfectly. In applied work, persistent poor fit, extreme concentration on one donor, or instability under small donor-pool changes are often signs that the treated unit is poorly supported by the donors.

Appendix S2: A formal potential-outcomes view. [Back to the synthetic-control estimand.](#)

Suppose treatment begins after T_0 and only unit 1 is treated. For post-treatment dates $t > T_0$, observed outcomes satisfy

$$Y_{1t} = Y_{1t}(1), \quad Y_{jt} = Y_{jt}(0) \text{ for } j = 2, \dots, J+1.$$

The target effect is

$$\tau_{1t} = Y_{1t}(1) - Y_{1t}(0).$$

Because $Y_{1t}(0)$ is unobserved after treatment, synthetic control approximates it by

$$\hat{Y}_{1t}(0) = \sum_{j=2}^{J+1} \hat{w}_j Y_{jt}.$$

This yields the estimator

$$\hat{\tau}_{1t} = Y_{1t} - \hat{Y}_{1t}(0).$$

The identifying content of the method is therefore concentrated in one empirical claim: the weighted donor combination chosen from pre-treatment information remains a credible approximation to the treated unit's untreated post-treatment path. The more convincingly the pre-treatment data and institutional context support that claim, the more persuasive the resulting treatment-effect path becomes.

Lecture 3 Summary

Lecture 3 adds a transparent small-treated-sample design to the panel toolkit:

- synthetic control constructs a weighted untreated comparison when no single control is convincing;
- the method is built around reconstructing the treated unit's untreated path from donor units;
- pre-treatment fit is the key diagnostic because it disciplines the counterfactual construction;
- placebo and sensitivity exercises are part of the design argument;
- the method is strongest when the donor pool is credible, the pre-treatment window is long, and the treated unit is well supported by the donor data.

Why this matters. Synthetic control extends the central theme of the course. Good econometrics is not a search for more complicated regressions. It is a search for more credible comparisons.

Worked Examples

Example 1: Pooled OLS, FE, and FE with Time Effects

This example will track how a coefficient changes as the specification moves from pooled OLS to unit FE and then to two-way FE. The interpretation will center on what source of variation is being removed at each step.

Example 2: Difference-in-Differences in Practice

This example will use a stylized policy reform to map treatment, control, timing, and identifying assumptions into a concrete regression setup. The discussion will stress both interpretation and empirical threats.

Summary Tables

Static Panel Estimators

This table will compare pooled OLS, FE, FD, and RE by identifying variation, assumptions, strengths, and weaknesses. Its purpose is to give students a compact decision map for standard panel settings.

Causal and Dynamic Panel Designs

This table will compare FE, DiD, and dynamic-panel estimators by research question, identification logic, and main risks. It will act as a bridge from foundational panel methods to modern applied practice.

Problem Set

Conceptual Questions

This subsection will contain short questions on identification, interpretation, and assumptions. The aim is to test economic reasoning before algebra.

Derivation Questions

This subsection will contain focused derivations on pooled-OLS bias, the within estimator, differencing, and DiD algebra. The questions will reinforce the mechanics needed for lecture discussion.

Applied Questions

This subsection will contain small empirical-design exercises asking students to diagnose estimands, threats, and appropriate standard errors. Each question will come with a brief solution sketch in the final version.

Unifying Perspective

This concluding section will tie together panel methods as different ways of isolating credible within-unit or policy-induced variation. It will also summarize how assumptions become stronger as the design moves from descriptive panel regressions to causal and dynamic settings.